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In [7]: # --- SAE-LAPE Romanization Analysis: Data Loading and Aggregation ---

import pandas as pd
from pathlib import Path
import numpy as np
import matplotlib.pyplot as plt

# === CONFIG ===
base_dir = "/home/aastha/multilingual_interpretability/identification/Llama-
experiment_root = "dakshina-{variant}-hi-sd-ur-bn/train"
languages = ["hi", "sd", "ur", "bn"]
num_layers = 16

# === LOAD DATA ===
sae_records = []

for variant in ["native", "romanized"]:
    for layer in range(num_layers):
        layer_dir = Path(base_dir) / f"layer_{layer}" / experiment_root
        for lang in languages:
            fpath = layer_dir / f"{lang}.csv"
            print(fpath)
            if not fpath.exists():
                continue
            df = pd.read_csv(fpath)
            print(df.head())
            df["variant"] = variant
            df["layer"] = layer
            df["language"] = lang
            sae_records.append(df)
print(sae_records)

df_sae = pd.concat(sae_records, ignore_index=True)
display(df_sae.head())

print(f"[INFO] Loaded {len(df_sae)} rows across {df_sae.layer.nunique()} lay
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/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_0/dakshina-native-hi-sd-ur-bn/train/hi.csv

	feature_idx	entropy	activation_prob	rank
0	31608	0.189624	0.187590	1
1	4073	0.194023	0.193167	2
2	105283	0.290827	0.344704	3
3	44340	0.363738	0.184050	4
4	35032	0.372147	0.177662	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_0/dakshina-native-hi-sd-ur-bn/train/sd.csv

	feature_idx	entropy	activation_prob	rank
0	16493	0.418848	0.208612	1
1	36886	0.463653	0.100328	2
2	24658	0.476556	0.381567	3
3	100458	0.484516	0.126389	4
4	65351	0.495110	0.120590	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_0/dakshina-native-hi-sd-ur-bn/train/ur.csv

	feature_idx	entropy	activation_prob	rank
0	5985	0.310431	0.145281	1
1	103404	0.316259	0.164240	2
2	45674	0.326929	0.232389	3
3	102209	0.350056	0.488958	4
4	102340	0.380485	0.174799	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_0/dakshina-native-hi-sd-ur-bn/train/bn.csv

	feature_idx	entropy	activation_prob	rank
0	95291	0.146979	0.242355	1
1	94542	0.247505	0.239887	2
2	44642	0.287293	0.311003	3
3	58968	0.291031	0.178009	4
4	92360	0.300357	0.602242	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_1/dakshina-native-hi-sd-ur-bn/train/hi.csv

	feature_idx	entropy	activation_prob	rank
0	96905	0.436213	0.224236	1
1	23452	0.441778	0.281809	2
2	64956	0.457899	0.161869	3
3	84670	0.512057	0.228086	4
4	35569	0.537851	0.681086	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_1/dakshina-native-hi-sd-ur-bn/train/sd.csv

	feature_idx	entropy	activation_prob	rank
0	123017	0.485620	0.268435	1
1	98581	0.585387	0.267788	2
2	2652	0.646580	0.400023	3
3	37903	0.732841	0.187307	4
4	71268	0.803699	0.627062	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_1/dakshina-native-hi-sd-ur-bn/train/ur.csv

	feature_idx	entropy	activation_prob	rank
0	78833	0.365825	0.227133	1
1	29250	0.378611	0.253910	2
2	88890	0.428253	0.355556	3
3	107543	0.454819	0.317362	4
4	126968	0.469483	0.119171	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_1/dakshina-native-hi-sd-ur-bn/train/bn.csv

	feature_idx	entropy	activation_prob	rank
0	2090	0.115268	0.445335	1
1	11517	0.137123	0.347889	2
2	47175	0.164478	0.203281	3
3	41039	0.166290	0.574320	4
4	21613	0.261686	0.214359	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_2/dakshina-native-hi-sd-ur-bn/train/hi.csv

	feature_idx	entropy	activation_prob	rank
0	5047	0.273816	0.132065	1
1	85181	0.358886	0.272085	2
2	127596	0.368599	0.645674	3
3	105685	0.376921	0.292604	4
4	48498	0.405206	0.546944	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_2/dakshina-native-hi-sd-ur-bn/train/sd.csv

	feature_idx	entropy	activation_prob	rank
0	63453	0.847335	0.284620	1
1	98877	0.907761	0.384813	2
2	30665	0.909846	0.629671	3
3	42294	0.950173	0.220613	4
4	117787	1.179023	0.231998	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_2/dakshina-native-hi-sd-ur-bn/train/ur.csv

	feature_idx	entropy	activation_prob	rank
0	130000	0.210658	0.192999	1
1	103610	0.446783	0.161192	2
2	37179	0.470038	0.194227	3
3	60454	0.481878	0.390585	4
4	96229	0.571093	0.164839	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_2/dakshina-native-hi-sd-ur-bn/train/bn.csv

	feature_idx	entropy	activation_prob	rank
0	13016	0.219705	0.667505	1
1	100818	0.316071	0.689119	2
2	122710	0.334815	0.194851	3
3	127955	0.349415	0.271740	4
4	90265	0.351225	0.271070	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_3/dakshina-native-hi-sd-ur-bn/train/hi.csv

	feature_idx	entropy	activation_prob	rank
0	58654	0.165214	0.299160	1
1	7745	0.249163	0.368465	2
2	23191	0.288734	0.137880	3
3	100927	0.297358	0.100170	4
4	99107	0.341021	0.127428	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_3/dakshina-native-hi-sd-ur-bn/train/sd.csv

	feature_idx	entropy	activation_prob	rank
0	85165	0.646525	0.219297	1
1	44718	0.873318	0.244841	2
2	39397	0.897794	0.228394	3
3	78906	1.034030	0.130697	4
4	102058	1.148329	0.136438	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_3/dakshina-native-hi-sd-ur-bn/train/ur.csv

	feature_idx	entropy	activation_prob	rank
0	19869	0.337185	0.396601	1
1	42492	0.369403	0.110525	2
2	14134	0.463071	0.106253	3
3	122402	0.472301	0.174860	4
4	31079	0.519560	0.402645	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_3/dakshina-native-hi-sd-ur-bn/train/bn.csv

	feature_idx	entropy	activation_prob	rank
0	58889	0.293560	0.171021	1
1	113544	0.393725	0.570196	2
2	74991	0.521463	0.164516	3
3	108370	0.572799	0.509789	4
4	56188	0.578856	0.116451	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_4/dakshina-native-hi-sd-ur-bn/train/hi.csv

	feature_idx	entropy	activation_prob	rank
0	42335	0.377120	0.136250	1
1	16104	0.378408	0.230927	2
2	88721	0.392424	0.139223	3
3	2201	0.417385	0.160181	4
4	107881	0.440311	0.110891	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_4/dakshina-native-hi-sd-ur-bn/train/sd.csv

	feature_idx	entropy	activation_prob	rank
0	7879	0.739179	0.147177	1
1	54586	0.902419	0.432384	2
2	60109	0.952479	0.228479	3
3	42936	1.009779	0.563210	4
4	27497	1.079323	0.527191	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_4/dakshina-native-hi-sd-ur-bn/train/ur.csv

	feature_idx	entropy	activation_prob	rank
0	11486	0.546145	0.147846	1
1	66006	0.840774	0.747299	2
2	126437	0.854812	0.159803	3
3	63461	1.008931	0.343473	4
4	73485	1.015106	0.251680	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_4/dakshina-native-hi-sd-ur-bn/train/bn.csv

	feature_idx	entropy	activation_prob	rank
0	118853	0.340786	0.103403	1
1	11770	0.454280	0.366985	2
2	81393	0.464026	0.128307	3
3	89150	0.477310	0.112091	4
4	125956	0.497448	0.239193	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_5/dakshina-native-hi-sd-ur-bn/train/hi.csv

	feature_idx	entropy	activation_prob	rank
0	4032	0.217586	0.172918	1
1	47139	0.253843	0.245701	2
2	25324	0.315684	0.354477	3
3	70326	0.371283	0.253641	4
4	18441	0.426557	0.323414	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_5/dakshina-native-hi-sd-ur-bn/train/sd.csv

	feature_idx	entropy	activation_prob	rank
0	68328	0.678062	0.129011	1
1	62076	0.685773	0.129606	2
2	67347	0.785117	0.382380	3
3	109519	0.916773	0.150454	4
4	45976	0.933526	0.281490	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_5/dakshina-native-hi-sd-ur-bn/train/ur.csv

	feature_idx	entropy	activation_prob	rank
0	22677	0.387299	0.187046	1
1	47325	0.481401	0.126986	2
2	126726	0.563512	0.177208	3
3	113393	0.630242	0.107197	4
4	41365	0.640313	0.494864	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_5/dakshina-native-hi-sd-ur-bn/train/bn.csv

	feature_idx	entropy	activation_prob	rank
0	119850	0.138607	0.386526	1
1	123915	0.339423	0.490011	2
2	128075	0.365026	0.220501	3
3	34876	0.366185	0.644064	4
4	68687	0.393938	0.272562	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_6/dakshina-native-hi-sd-ur-bn/train/hi.csv

	feature_idx	entropy	activation_prob	rank
0	17446	0.536685	0.341892	1
1	121904	0.549588	0.144329	2
2	42561	0.575698	0.193306	3
3	85084	0.597224	0.321976	4
4	88193	0.598651	0.194706	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_6/dakshina-native-hi-sd-ur-bn/train/sd.csv

	feature_idx	entropy	activation_prob	rank
0	102053	0.805024	0.191561	1
1	76881	0.807733	0.373810	2
2	72786	0.840375	0.294893	3
3	6750	0.911501	0.474884	4
4	87999	0.917313	0.537743	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_6/dakshina-native-hi-sd-ur-bn/train/ur.csv

	feature_idx	entropy	activation_prob	rank
0	107504	0.450361	0.205923	1
1	8310	0.695308	0.266270	2
2	100886	0.747418	0.133747	3
3	126999	0.761011	0.365063	4
4	82694	0.798017	0.443033	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_6/dakshina-native-hi-sd-ur-bn/train/bn.csv

	feature_idx	entropy	activation_prob	rank
0	60839	0.064536	0.173018	1
1	100724	0.201433	0.420769	2
2	18544	0.255511	0.366203	3
3	118884	0.266163	0.301128	4
4	14421	0.278066	0.534946	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_7/dakshina-native-hi-sd-ur-bn/train/hi.csv

	feature_idx	entropy	activation_prob	rank
0	127009	0.244622	0.250130	1
1	71743	0.245703	0.150795	2
2	108375	0.276613	0.308105	3
3	6083	0.287308	0.108370	4
4	114405	0.309442	0.351782	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_7/dakshina-native-hi-sd-ur-bn/train/sd.csv

	feature_idx	entropy	activation_prob	rank
0	37822	0.699503	0.204899	1
1	14028	0.763669	0.144428	2
2	106783	0.886693	0.330263	3
3	3014	0.895475	0.167557	4
4	90836	0.934217	0.139751	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_7/dakshina-native-hi-sd-ur-bn/train/ur.csv

	feature_idx	entropy	activation_prob	rank
0	16878	0.443546	0.107597	1
1	39232	0.518938	0.410769	2
2	126750	0.668657	0.310461	3
3	123657	0.684976	0.191267	4
4	123742	0.692506	0.207653	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_7/dakshina-native-hi-sd-ur-bn/train/bn.csv

	feature_idx	entropy	activation_prob	rank
0	10445	0.115132	0.534714	1
1	9559	0.172530	0.649775	2
2	120313	0.178084	0.589148	3
3	32350	0.180815	0.290201	4
4	63385	0.225100	0.672256	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_8/dakshina-native-hi-sd-ur-bn/train/hi.csv

	feature_idx	entropy	activation_prob	rank
0	95105	0.287682	0.117023	1
1	70585	0.292696	0.106428	2
2	94301	0.308818	0.253723	3
3	81439	0.386021	0.474476	4
4	42565	0.415311	0.167820	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_8/dakshina-native-hi-sd-ur-bn/train/sd.csv

	feature_idx	entropy	activation_prob	rank
0	82124	0.645793	0.113132	1
1	82686	0.660742	0.300192	2
2	120590	0.730823	0.347992	3
3	62940	0.765307	0.231215	4
4	18513	0.783665	0.202015	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_8/dakshina-native-hi-sd-ur-bn/train/ur.csv

	feature_idx	entropy	activation_prob	rank
0	103407	0.327939	0.249063	1
1	57930	0.462347	0.141731	2
2	18578	0.577233	0.369690	3
3	93636	0.626865	0.115406	4
4	13658	0.695858	0.263330	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_8/dakshina-native-hi-sd-ur-bn/train/bn.csv

	feature_idx	entropy	activation_prob	rank
0	46515	0.307537	0.546928	1
1	130928	0.365256	0.168512	2
2	45221	0.436766	0.111017	3
3	87763	0.447049	0.118967	4
4	71216	0.460818	0.496729	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_9/dakshina-native-hi-sd-ur-bn/train/hi.csv

	feature_idx	entropy	activation_prob	rank
0	61493	0.360224	0.223667	1
1	63798	0.392165	0.388933	2
2	62908	0.437122	0.152121	3
3	54578	0.441578	0.328499	4
4	101730	0.461004	0.266512	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_9/dakshina-native-hi-sd-ur-bn/train/sd.csv

	feature_idx	entropy	activation_prob	rank
0	123229	0.608654	0.197311	1
1	74376	0.724760	0.263445	2
2	116401	0.829822	0.640194	3
3	38337	0.833839	0.153450	4
4	69469	0.841954	0.388235	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_9/dakshina-native-hi-sd-ur-bn/train/ur.csv

	feature_idx	entropy	activation_prob	rank
0	11752	0.629756	0.284811	1
1	21757	0.750072	0.150667	2
2	41193	0.789260	0.328083	3
3	51107	0.830868	0.415403	4
4	110707	0.842732	0.368752	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_9/dakshina-native-hi-sd-ur-bn/train/bn.csv

	feature_idx	entropy	activation_prob	rank
0	105205	0.130511	0.620911	1
1	14634	0.367555	0.320236	2
2	128315	0.383841	0.618185	3
3	119625	0.560913	0.439518	4
4	113107	0.608316	0.730923	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_10/dakshina-native-hi-sd-ur-bn/train/hi.csv

	feature_idx	entropy	activation_prob	rank
0	88307	0.241521	0.152971	1
1	126404	0.261781	0.160624	2
2	104804	0.320593	0.149509	3
3	60535	0.445323	0.115124	4
4	113353	0.505494	0.238970	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_10/dakshina-native-hi-sd-ur-bn/train/sd.csv

	feature_idx	entropy	activation_prob	rank
0	93613	0.751195	0.532440	1
1	80967	0.792047	0.170916	2
2	51243	0.813049	0.450006	3
3	126164	0.819055	0.209738	4
4	42928	0.917367	0.464216	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_10/dakshina-native-hi-sd-ur-bn/train/ur.csv

	feature_idx	entropy	activation_prob	rank
0	79021	0.590797	0.390512	1
1	51962	0.595602	0.386360	2
2	101038	0.657552	0.191652	3
3	3070	0.687200	0.144587	4
4	109779	0.727153	0.105375	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_10/dakshina-native-hi-sd-ur-bn/train/bn.csv

	feature_idx	entropy	activation_prob	rank
0	59389	0.161388	0.626919	1
1	120013	0.177218	0.503796	2
2	89912	0.250207	0.461046	3
3	19487	0.271630	0.285940	4
4	22877	0.277049	0.137423	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_11/dakshina-native-hi-sd-ur-bn/train/hi.csv

	feature_idx	entropy	activation_prob	rank
0	58797	0.176933	0.234520	1
1	94798	0.247413	0.233931	2
2	110916	0.267219	0.158310	3
3	106154	0.274694	0.245832	4
4	12505	0.300747	0.277353	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_11/dakshina-native-hi-sd-ur-bn/train/sd.csv

	feature_idx	entropy	activation_prob	rank
0	127934	0.560810	0.174768	1
1	29415	0.874438	0.259378	2
2	95067	0.944895	0.484579	3
3	28418	1.025589	0.426176	4
4	119400	1.046538	0.477745	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_11/dakshina-native-hi-sd-ur-bn/train/ur.csv

	feature_idx	entropy	activation_prob	rank
0	92824	0.302455	0.103971	1
1	102107	0.413422	0.172382	2
2	45072	0.417631	0.140699	3
3	124259	0.512953	0.158079	4
4	45466	0.591103	0.399515	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_11/dakshina-native-hi-sd-ur-bn/train/bn.csv

	feature_idx	entropy	activation_prob	rank
0	67261	0.089749	0.164030	1
1	25098	0.166364	0.289244	2
2	28176	0.170144	0.542049	3
3	23702	0.181190	0.292145	4
4	69332	0.196614	0.237854	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_12/dakshina-native-hi-sd-ur-bn/train/hi.csv

	feature_idx	entropy	activation_prob	rank
0	81687	0.302307	0.156092	1
1	90837	0.308213	0.218586	2
2	9034	0.462300	0.190767	3
3	123118	0.636046	0.303753	4
4	4781	0.636771	0.347107	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_12/dakshina-native-hi-sd-ur-bn/train/sd.csv

	feature_idx	entropy	activation_prob	rank
0	47175	1.083773	0.321817	1
1	106812	1.181344	0.414875	2

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_12/dakshina-native-hi-sd-ur-bn/train/ur.csv

	feature_idx	entropy	activation_prob	rank
0	34314	0.336195	0.280132	1
1	121099	0.717466	0.608160	2
2	11443	0.888802	0.424414	3
3	35977	0.974853	0.649220	4
4	64470	1.021368	0.343220	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_12/dakshina-native-hi-sd-ur-bn/train/bn.csv

	feature_idx	entropy	activation_prob	rank
0	67845	0.559479	0.341810	1
1	110041	0.660086	0.259180	2
2	89373	0.660421	0.143581	3
3	130435	0.669846	0.419643	4
4	17074	0.696286	0.161865	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_13/dakshina-native-hi-sd-ur-bn/train/hi.csv

	feature_idx	entropy	activation_prob	rank
0	17535	0.478949	0.170988	1
1	31632	0.528341	0.187696	2
2	40379	0.534223	0.193819	3
3	75200	0.542113	0.176958	4
4	75374	0.581041	0.134613	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_13/dakshina-native-hi-sd-ur-bn/train/sd.csv

	feature_idx	entropy	activation_prob	rank
0	101492	0.910673	0.250774	1
1	73338	1.252215	0.253017	2

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_13/dakshina-native-hi-sd-ur-bn/train/ur.csv

	feature_idx	entropy	activation_prob	rank
0	82277	0.630004	0.363119	1
1	11303	0.823517	0.261728	2
2	36160	0.874833	0.475115	3
3	25265	1.046269	0.251162	4
4	115249	1.145033	0.121283	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_13/dakshina-native-hi-sd-ur-bn/train/bn.csv

	feature_idx	entropy	activation_prob	rank
0	120470	0.100562	0.358427	1
1	103214	0.110556	0.368982	2
2	22650	0.159867	0.733921	3
3	82433	0.176703	0.265450	4
4	102791	0.176896	0.177736	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_14/dakshina-native-hi-sd-ur-bn/train/hi.csv

	feature_idx	entropy	activation_prob	rank
0	93319	0.413439	0.152750	1
1	31396	0.494020	0.148100	2
2	122631	0.540788	0.106525	3

3	56311	0.552064	0.334877	4
4	87335	0.552554	0.744989	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_14/dakshina-native-hi-sd-ur-bn/train/sd.csv

	feature_idx	entropy	activation_prob	rank
0	124424	0.397374	0.125715	1
1	80083	0.831855	0.346664	2
2	118728	0.849559	0.171066	3
3	13227	0.914345	0.469572	4
4	48363	0.951757	0.113306	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_14/dakshina-native-hi-sd-ur-bn/train/ur.csv

	feature_idx	entropy	activation_prob	rank
0	116413	0.535478	0.460991	1
1	75696	0.545065	0.121029	2
2	51359	0.654343	0.178436	3
3	61288	0.659408	0.376837	4
4	77841	0.692818	0.402769	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_14/dakshina-native-hi-sd-ur-bn/train/bn.csv

	feature_idx	entropy	activation_prob	rank
0	18521	0.114858	0.157995	1
1	50516	0.201793	0.174479	2
2	85094	0.202823	0.156498	3
3	45439	0.215060	0.163494	4
4	21121	0.247429	0.158327	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_15/dakshina-native-hi-sd-ur-bn/train/hi.csv

	feature_idx	entropy	activation_prob	rank
0	100144	0.740426	0.338015	1
1	86589	0.791592	0.264290	2
2	15283	0.820698	0.213249	3
3	44430	0.857847	0.413305	4
4	108004	0.919114	0.519995	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_15/dakshina-native-hi-sd-ur-bn/train/sd.csv

	feature_idx	entropy	activation_prob	rank
0	94601	1.24385	0.685205	1

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_15/dakshina-native-hi-sd-ur-bn/train/ur.csv

	feature_idx	entropy	activation_prob	rank
0	125945	0.98112	0.245409	1

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_15/dakshina-native-hi-sd-ur-bn/train/bn.csv

	feature_idx	entropy	activation_prob	rank
0	110312	0.821208	0.186544	1
1	72534	1.187816	0.101597	2
2	93168	1.210266	0.325003	3
3	75805	1.222122	0.116377	4
4	95412	1.223444	0.102938	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_0/dakshina-romanized-hi-sd-ur-bn/train/hi.csv

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_0/dakshina-romanized-hi-sd-ur-bn/train/sd.csv

	feature_idx	entropy	activation_prob	rank
0	130721	1.231083	0.109940	1

1	35986	1.245613	0.353111	2
2	126470	1.267289	0.113033	3

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_0/dakshina-romanized-hi-sd-ur-bn/train/ur.csv

	feature_idx	entropy	activation_prob	rank
0	124209	1.175587	0.120884	1
1	73545	1.252441	0.163622	2
2	98978	1.255928	0.158600	3
3	57639	1.289057	0.115953	4
4	66723	1.297242	0.118437	5

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_0/dakshina-romanized-hi-sd-ur-bn/train/bn.csv

	feature_idx	entropy	activation_prob	rank
0	76571	1.299738	0.220369	1

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_1/dakshina-romanized-hi-sd-ur-bn/train/hi.csv

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_1/dakshina-romanized-hi-sd-ur-bn/train/sd.csv

	feature_idx	entropy	activation_prob	rank
0	105661	1.148494	0.132753	1

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_1/dakshina-romanized-hi-sd-ur-bn/train/ur.csv

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_1/dakshina-romanized-hi-sd-ur-bn/train/bn.csv

	feature_idx	entropy	activation_prob	rank
0	2382	1.256221	0.299067	1

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_2/dakshina-romanized-hi-sd-ur-bn/train/hi.csv

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_2/dakshina-romanized-hi-sd-ur-bn/train/sd.csv

	feature_idx	entropy	activation_prob	rank
0	105811	1.218724	0.141397	1

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_2/dakshina-romanized-hi-sd-ur-bn/train/ur.csv

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_2/dakshina-romanized-hi-sd-ur-bn/train/bn.csv

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_3/dakshina-romanized-hi-sd-ur-bn/train/hi.csv

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_3/dakshina-romanized-hi-sd-ur-bn/train/sd.csv

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_3/dakshina-romanized-hi-sd-ur-bn/train/ur.csv

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_3/dakshina-romanized-hi-sd-ur-bn/train/bn.csv

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_4/dakshina-romanized-hi-sd-ur-bn/train/hi.csv

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_4/dakshina-romanized-hi-sd-ur-bn/train/sd.csv

	feature_idx	entropy	activation_prob	rank
0	1028	1.153829	0.115709	1
1	30384	1.255921	0.195217	2

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_4/dakshina-romanized-hi-sd-ur-bn/train/ur.csv

	feature_idx	entropy	activation_prob	rank
0	110146	1.224434	0.107779	1
1	56551	1.253462	0.111492	2

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/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_4/dakshina-romanized-hi-sd-ur-bn/train/bn.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_5/dakshina-romanized-hi-sd-ur-bn/train/hi.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_5/dakshina-romanized-hi-sd-ur-bn/train/sd.csv
    feature_idx  entropy  activation_prob  rank
0           45025  1.234088         0.101013    1
1           30198  1.266937         0.269325    2
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_5/dakshina-romanized-hi-sd-ur-bn/train/ur.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_5/dakshina-romanized-hi-sd-ur-bn/train/bn.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_6/dakshina-romanized-hi-sd-ur-bn/train/hi.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_6/dakshina-romanized-hi-sd-ur-bn/train/sd.csv
    feature_idx  entropy  activation_prob  rank
0           43429  1.229118         0.104574    1
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_6/dakshina-romanized-hi-sd-ur-bn/train/ur.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_6/dakshina-romanized-hi-sd-ur-bn/train/bn.csv
    feature_idx  entropy  activation_prob  rank
0           130328  1.247702         0.347742    1
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_7/dakshina-romanized-hi-sd-ur-bn/train/hi.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_7/dakshina-romanized-hi-sd-ur-bn/train/sd.csv
    feature_idx  entropy  activation_prob  rank
0           119935  1.165310         0.124025    1
1           54154  1.234290         0.164255    2
2           21015  1.250724         0.118206    3
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_7/dakshina-romanized-hi-sd-ur-bn/train/ur.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_7/dakshina-romanized-hi-sd-ur-bn/train/bn.csv
    feature_idx  entropy  activation_prob  rank
0            827  1.285219         0.399790    1
1           45382  1.295356         0.269762    2
2           115854  1.310983         0.584940    3
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_8/dakshina-romanized-hi-sd-ur-bn/train/hi.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_8/dakshina-romanized-hi-sd-ur-bn/train/sd.csv
    feature_idx  entropy  activation_prob  rank
0           86682  1.146242         0.114645    1
1           32443  1.255433         0.179714    2
2           119100  1.292345         0.169138    3
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_8/dakshina-romanized-hi-sd-ur-bn/train/ur.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_8/dakshina-romanized-hi-sd-ur-bn/train/bn.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_9/dakshina-romanized-hi-sd-ur-bn/train/hi.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l

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ape/layer_9/dakshina-romanized-hi-sd-ur-bn/train/sd.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_9/dakshina-romanized-hi-sd-ur-bn/train/ur.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_9/dakshina-romanized-hi-sd-ur-bn/train/bn.csv
    feature_idx  entropy  activation_prob  rank
0         116401  1.282878         0.463554      1
1         80474  1.315335         0.734299      2
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_10/dakshina-romanized-hi-sd-ur-bn/train/hi.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_10/dakshina-romanized-hi-sd-ur-bn/train/sd.csv
    feature_idx  entropy  activation_prob  rank
0         74599  1.069244         0.187508      1
1         37493  1.308289         0.231314      2
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_10/dakshina-romanized-hi-sd-ur-bn/train/ur.csv
    feature_idx  entropy  activation_prob  rank
0         109779  1.134843         0.137804      1
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_10/dakshina-romanized-hi-sd-ur-bn/train/bn.csv
    feature_idx  entropy  activation_prob  rank
0         13811  1.274303         0.224667      1
1         47356  1.297246         0.240349      2
2         110774  1.316236         0.208357      3
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_11/dakshina-romanized-hi-sd-ur-bn/train/hi.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_11/dakshina-romanized-hi-sd-ur-bn/train/sd.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_11/dakshina-romanized-hi-sd-ur-bn/train/ur.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_11/dakshina-romanized-hi-sd-ur-bn/train/bn.csv
    feature_idx  entropy  activation_prob  rank
0         101304  1.272928         0.333155      1
1         61455  1.295832         0.294790      2
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_12/dakshina-romanized-hi-sd-ur-bn/train/hi.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_12/dakshina-romanized-hi-sd-ur-bn/train/sd.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_12/dakshina-romanized-hi-sd-ur-bn/train/ur.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_12/dakshina-romanized-hi-sd-ur-bn/train/bn.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_13/dakshina-romanized-hi-sd-ur-bn/train/hi.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_13/dakshina-romanized-hi-sd-ur-bn/train/sd.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_13/dakshina-romanized-hi-sd-ur-bn/train/ur.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_13/dakshina-romanized-hi-sd-ur-bn/train/bn.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_14/dakshina-romanized-hi-sd-ur-bn/train/hi.csv
/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_l
ape/layer_14/dakshina-romanized-hi-sd-ur-bn/train/sd.csv

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	feature_idx	entropy	activation_prob	rank
0	55963	1.299686	0.257774	1

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_14/dakshina-romanized-hi-sd-ur-bn/train/ur.csv

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_14/dakshina-romanized-hi-sd-ur-bn/train/bn.csv

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_15/dakshina-romanized-hi-sd-ur-bn/train/hi.csv

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_15/dakshina-romanized-hi-sd-ur-bn/train/sd.csv

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_15/dakshina-romanized-hi-sd-ur-bn/train/ur.csv

/home/aastha/multilingual_interpretability/identification/Llama-3.2-1B/sae_lape/layer_15/dakshina-romanized-hi-sd-ur-bn/train/bn.csv

	feature_idx	entropy	activation_prob	rank	variant	layer	language
0	31608	0.189624	0.187590	1	native	0	hi
1	4073	0.194023	0.193167	2	native	0	hi
2	105283	0.290827	0.344704	3	native	0	hi
3	44340	0.363738	0.184050	4	native	0	hi
4	35032	0.372147	0.177662	5	native	0	hi
..	...	...	...	...	...	...	...
95	2406	0.756674	0.221430	96	native	0	hi
96	129851	0.759914	0.679062	97	native	0	hi
97	58510	0.760956	0.360678	98	native	0	hi
98	35887	0.762351	0.731468	99	native	0	hi
99	66041	0.766101	0.509464	100	native	0	hi

[100 rows x 7 columns],

	feature_idx	entropy	activation_prob	rank	variant	layer	language
0	16493	0.418848	0.208612	1	native	0	sd
1	36886	0.463653	0.100328	2	native	0	sd
2	24658	0.476556	0.381567	3	native	0	sd
3	100458	0.484516	0.126389	4	native	0	sd
4	65351	0.495110	0.120590	5	native	0	sd
..	...	...	...	...	...	...	...
62	75534	1.257684	0.407814	63	native	0	sd
63	30163	1.260599	0.267733	64	native	0	sd
64	124168	1.261395	0.544640	65	native	0	sd
65	65044	1.267572	0.434616	66	native	0	sd
66	30787	1.300347	0.268280	67	native	0	sd

[67 rows x 7 columns],

	feature_idx	entropy	activation_prob	rank	variant	layer	language
0	5985	0.310431	0.145281	1	native	0	ur
1	103404	0.316259	0.164240	2	native	0	ur
2	45674	0.326929	0.232389	3	native	0	ur
3	102209	0.350056	0.488958	4	native	0	ur
4	102340	0.380485	0.174799	5	native	0	ur
..	...	...	...	...	...	...	...
95	110910	1.072703	0.204201	96	native	0	ur
96	102348	1.086228	0.123550	97	native	0	ur
97	57639	1.087633	0.265919	98	native	0	ur
98	95868	1.102047	0.103819	99	native	0	ur
99	97223	1.119107	0.450062	100	native	0	ur

[100 rows x 7 columns],

	feature_idx	entropy	activation_prob	rank	variant
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variant	layer	language	feature_idx	entropy	activation_prob	rank	var
0	95291	0.146979	0.242355	1	native	0	bn
1	94542	0.247505	0.239887	2	native	0	bn
2	44642	0.287293	0.311003	3	native	0	bn
3	58968	0.291031	0.178009	4	native	0	bn
4	92360	0.300357	0.602242	5	native	0	bn
..	...	...	...	...	...	...	...
95	34503	1.058443	0.120403	96	native	0	bn
96	85124	1.064042	0.112993	97	native	0	bn
97	30907	1.065102	0.630718	98	native	0	bn
98	41448	1.068350	0.269157	99	native	0	bn
99	93585	1.072706	0.462193	100	native	0	bn

[100 rows x 7 columns],

variant	layer	language	feature_idx	entropy	activation_prob	rank	var
0	96905	0.436213	0.224236	1	native	1	hi
1	23452	0.441778	0.281809	2	native	1	hi
2	64956	0.457899	0.161869	3	native	1	hi
3	84670	0.512057	0.228086	4	native	1	hi
4	35569	0.537851	0.681086	5	native	1	hi
..	...	...	...	...	...	...	...
95	29117	0.905275	0.364336	96	native	1	hi
96	103601	0.905918	0.263859	97	native	1	hi
97	90809	0.906363	0.180714	98	native	1	hi
98	22168	0.907049	0.418484	99	native	1	hi
99	130167	0.907978	0.371912	100	native	1	hi

[100 rows x 7 columns],

variant	layer	language	feature_idx	entropy	activation_prob	rank	var
0	123017	0.485620	0.268435	1	native	1	sd
1	98581	0.585387	0.267788	2	native	1	sd
2	2652	0.646580	0.400023	3	native	1	sd
3	37903	0.732841	0.187307	4	native	1	sd
4	71268	0.803699	0.627062	5	native	1	sd
5	4501	0.819372	0.301971	6	native	1	sd
6	40785	0.832488	0.357637	7	native	1	sd
7	57902	0.847770	0.396980	8	native	1	sd
8	80716	0.870437	0.169497	9	native	1	sd
9	32234	0.875460	0.541233	10	native	1	sd
10	44379	0.897315	0.185236	11	native	1	sd
11	128693	0.920118	0.144953	12	native	1	sd
12	91023	0.929021	0.100397	13	native	1	sd
13	90332	0.958959	0.302391	14	native	1	sd
14	99999	0.962268	0.289330	15	native	1	sd
15	120402	1.011496	0.156685	16	native	1	sd
16	55536	1.029502	0.211496	17	native	1	sd
17	19756	1.033186	0.164852	18	native	1	sd
18	14865	1.049663	0.311699	19	native	1	sd
19	1823	1.097533	0.151783	20	native	1	sd
20	67165	1.106059	0.528181	21	native	1	sd
21	15943	1.164730	0.151840	22	native	1	sd
22	79630	1.178511	0.117708	23	native	1	sd
23	104586	1.250724	0.765417	24	native	1	sd,

feature_idx	entropy	activation_prob	rank	variant	layer	language	
0	78833	0.365825	0.227133	1	native	1	ur
1	29250	0.378611	0.253910	2	native	1	ur

2	88890	0.428253	0.355556	3	native	1	ur
3	107543	0.454819	0.317362	4	native	1	ur
4	126968	0.469483	0.119171	5	native	1	ur
..	...	...	...	...	...	...	...
80	106808	1.229741	0.252337	81	native	1	ur
81	96896	1.262011	0.294039	82	native	1	ur
82	93306	1.286080	0.491682	83	native	1	ur
83	26592	1.290158	0.233572	84	native	1	ur
84	65633	1.308675	0.535460	85	native	1	ur

[85 rows x 7 columns],							
	feature_idx	entropy	activation_prob	rank	variant	layer	language
0	2090	0.115268	0.445335	1	native	1	bn
1	11517	0.137123	0.347889	2	native	1	bn
2	47175	0.164478	0.203281	3	native	1	bn
3	41039	0.166290	0.574320	4	native	1	bn
4	21613	0.261686	0.214359	5	native	1	bn
..	...	...	...	...	...	...	...
95	43282	0.734030	0.255835	96	native	1	bn
96	130375	0.736327	0.686941	97	native	1	bn
97	62173	0.737121	0.436932	98	native	1	bn
98	128303	0.737819	0.441864	99	native	1	bn
99	13265	0.743418	0.155573	100	native	1	bn

[100 rows x 7 columns],							
	feature_idx	entropy	activation_prob	rank	variant	layer	language
0	5047	0.273816	0.132065	1	native	2	hi
1	85181	0.358886	0.272085	2	native	2	hi
2	127596	0.368599	0.645674	3	native	2	hi
3	105685	0.376921	0.292604	4	native	2	hi
4	48498	0.405206	0.546944	5	native	2	hi
..	...	...	...	...	...	...	...
95	75006	0.898181	0.657906	96	native	2	hi
96	101452	0.899380	0.128485	97	native	2	hi
97	128429	0.901383	0.355993	98	native	2	hi
98	82585	0.901799	0.313507	99	native	2	hi
99	71931	0.910121	0.288090	100	native	2	hi

[100 rows x 7 columns],							
	feature_idx	entropy	activation_prob	rank	variant	layer	language
0	63453	0.847335	0.284620	1	native	2	sd
1	98877	0.907761	0.384813	2	native	2	sd
2	30665	0.909846	0.629671	3	native	2	sd
3	42294	0.950173	0.220613	4	native	2	sd
4	117787	1.179023	0.231998	5	native	2	sd
5	8081	1.222015	0.640596	6	native	2	sd
6	122088	1.227039	0.214579	7	native	2	sd
7	26659	1.272096	0.370457	8	native	2	sd,
feature_idx entropy activation_prob rank variant layer language							
0	130000	0.210658	0.192999	1	native	2	ur
1	103610	0.446783	0.161192	2	native	2	ur
2	37179	0.470038	0.194227	3	native	2	ur
3	60454	0.481878	0.390585	4	native	2	ur
4	96229	0.571093	0.164839	5	native	2	ur
5	53461	0.576236	0.472858	6	native	2	ur
6	25219	0.587531	0.304628	7	native	2	ur

7	89008	0.597016	0.221830	8	native	2	ur
8	87348	0.601535	0.486586	9	native	2	ur
9	24087	0.636240	0.240011	10	native	2	ur
10	45260	0.648056	0.385677	11	native	2	ur
11	3753	0.713865	0.190830	12	native	2	ur
12	57425	0.715065	0.378050	13	native	2	ur
13	37936	0.715528	0.129147	14	native	2	ur
14	82816	0.736578	0.694321	15	native	2	ur
15	70332	0.747176	0.223772	16	native	2	ur
16	77487	0.754035	0.292713	17	native	2	ur
17	2986	0.764742	0.584969	18	native	2	ur
18	100624	0.780223	0.141502	19	native	2	ur
19	89984	0.815767	0.117554	20	native	2	ur
20	2442	0.832568	0.468962	21	native	2	ur
21	68484	0.883130	0.106934	22	native	2	ur
22	87736	0.884894	0.585526	23	native	2	ur
23	21567	0.887454	0.640057	24	native	2	ur
24	28468	0.920862	0.374582	25	native	2	ur
25	87655	0.936246	0.755857	26	native	2	ur
26	87387	0.964664	0.448750	27	native	2	ur
27	23318	0.970274	0.126770	28	native	2	ur
28	93078	0.978766	0.345591	29	native	2	ur
29	18516	0.981126	0.658072	30	native	2	ur
30	125431	1.012385	0.289883	31	native	2	ur
31	62003	1.014682	0.214722	32	native	2	ur
32	102893	1.040642	0.185760	33	native	2	ur
33	73775	1.042111	0.128220	34	native	2	ur
34	20568	1.056194	0.342957	35	native	2	ur
35	81949	1.067136	0.466211	36	native	2	ur
36	23280	1.072586	0.606606	37	native	2	ur
37	22156	1.079459	0.336668	38	native	2	ur
38	8030	1.107878	0.116787	39	native	2	ur
39	111799	1.115020	0.153452	40	native	2	ur
40	84499	1.128319	0.140030	41	native	2	ur
41	94252	1.128967	0.256790	42	native	2	ur
42	12718	1.185879	0.628685	43	native	2	ur
43	121745	1.188783	0.709227	44	native	2	ur
44	82843	1.212674	0.454815	45	native	2	ur
45	32809	1.223910	0.111300	46	native	2	ur
46	35683	1.232050	0.681709	47	native	2	ur
47	63214	1.232201	0.254116	48	native	2	ur
48	93642	1.240912	0.261282	49	native	2	ur
49	107274	1.257819	0.212828	50	native	2	ur
50	24626	1.272155	0.310631	51	native	2	ur
51	78959	1.295621	0.254499	52	native	2	ur
52	97425	1.297456	0.228939	53	native	2	ur,
feature_idx entropy activation_prob rank variant layer language							
0	13016	0.219705	0.667505	1	native	2	bn
1	100818	0.316071	0.689119	2	native	2	bn
2	122710	0.334815	0.194851	3	native	2	bn
3	127955	0.349415	0.271740	4	native	2	bn
4	90265	0.351225	0.271070	5	native	2	bn
..	...	...	...	...	...	...	...
95	73045	0.859230	0.146496	96	native	2	bn
96	80704	0.862627	0.128281	97	native	2	bn
97	18546	0.867558	0.782221	98	native	2	bn

98	25422	0.867784	0.539293	99	native	2	bn
99	3213	0.868365	0.261078	100	native	2	bn

[100 rows x 7 columns],			feature_idx	entropy	activation_prob	rank	variant
layer	language						
0	58654	0.165214	0.299160	1	native	3	hi
1	7745	0.249163	0.368465	2	native	3	hi
2	23191	0.288734	0.137880	3	native	3	hi
3	100927	0.297358	0.100170	4	native	3	hi
4	99107	0.341021	0.127428	5	native	3	hi
..	...	...	...	...	...	...	...
95	112962	0.772675	0.354712	96	native	3	hi
96	78791	0.773688	0.223364	97	native	3	hi
97	109609	0.774733	0.300436	98	native	3	hi
98	128118	0.779137	0.214981	99	native	3	hi
99	66244	0.779559	0.358789	100	native	3	hi

[100 rows x 7 columns],			feature_idx	entropy	activation_prob	rank	variant
layer	language						
0	85165	0.646525	0.219297	1	native	3	sd
1	44718	0.873318	0.244841	2	native	3	sd
2	39397	0.897794	0.228394	3	native	3	sd
3	78906	1.034030	0.130697	4	native	3	sd
4	102058	1.148329	0.136438	5	native	3	sd
5	17591	1.170378	0.654491	6	native	3	sd
6	113793	1.210586	0.218857	7	native	3	sd
7	25922	1.240626	0.400240	8	native	3	sd
8	40885	1.283343	0.405502	9	native	3	sd,

feature_idx	entropy	activation_prob	rank	variant	layer	language
0	19869	0.337185	0.396601	1	native	3
1	42492	0.369403	0.110525	2	native	3
2	14134	0.463071	0.106253	3	native	3
3	122402	0.472301	0.174860	4	native	3
4	31079	0.519560	0.402645	5	native	3
5	96469	0.522978	0.579275	6	native	3
6	32769	0.626090	0.124630	7	native	3
7	125721	0.632562	0.122425	8	native	3
8	59112	0.697797	0.163296	9	native	3
9	13019	0.711014	0.434393	10	native	3
10	11313	0.711973	0.347954	11	native	3
11	84483	0.719375	0.147686	12	native	3
12	46859	0.730593	0.405319	13	native	3
13	103875	0.763038	0.713100	14	native	3
14	13529	0.779227	0.161865	15	native	3
15	117459	0.805668	0.232893	16	native	3
16	121275	0.838304	0.212523	17	native	3
17	59179	0.845104	0.518177	18	native	3
18	72420	0.855159	0.171999	19	native	3
19	85786	0.945082	0.332336	20	native	3
20	41580	0.951273	0.298780	21	native	3
21	99244	0.975953	0.582584	22	native	3
22	113002	0.988199	0.224016	23	native	3
23	51779	1.026033	0.168787	24	native	3
24	59420	1.049415	0.199465	25	native	3
25	120338	1.055386	0.581877	26	native	3
26	47830	1.060191	0.745398	27	native	3

27	89169	1.082042	0.425606	28	native	3	ur
28	29710	1.083944	0.452221	29	native	3	ur
29	36720	1.095161	0.480911	30	native	3	ur
30	122142	1.133159	0.313927	31	native	3	ur
31	18506	1.135627	0.653955	32	native	3	ur
32	62637	1.142389	0.499992	33	native	3	ur
33	112861	1.151429	0.109972	34	native	3	ur
34	67910	1.202656	0.864420	35	native	3	ur
35	123400	1.224640	0.101229	36	native	3	ur
36	11761	1.269579	0.123220	37	native	3	ur,

feature_idx	entropy	activation_prob	rank	variant	layer	language
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0	58889	0.293560	0.171021	1	native	3	bn
1	113544	0.393725	0.570196	2	native	3	bn
2	74991	0.521463	0.164516	3	native	3	bn
3	108370	0.572799	0.509789	4	native	3	bn
4	56188	0.578856	0.116451	5	native	3	bn
..	...	...	...	...	...	...	...
60	107414	1.200274	0.119448	61	native	3	bn
61	38975	1.265310	0.192232	62	native	3	bn
62	13714	1.270904	0.102959	63	native	3	bn
63	13649	1.290017	0.561010	64	native	3	bn
64	56088	1.300052	0.122694	65	native	3	bn

[65 rows x 7 columns],	feature_idx	entropy	activation_prob	rank	variant	layer	language
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0	42335	0.377120	0.136250	1	native	4	hi
1	16104	0.378408	0.230927	2	native	4	hi
2	88721	0.392424	0.139223	3	native	4	hi
3	2201	0.417385	0.160181	4	native	4	hi
4	107881	0.440311	0.110891	5	native	4	hi
..	...	...	...	...	...	...	...
95	111825	0.963339	0.805161	96	native	4	hi
96	84747	0.964424	0.695332	97	native	4	hi
97	69218	0.965342	0.416517	98	native	4	hi
98	92044	0.968334	0.488180	99	native	4	hi
99	75731	0.969060	0.355282	100	native	4	hi

[100 rows x 7 columns],	feature_idx	entropy	activation_prob	rank	variant	layer	language
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0	7879	0.739179	0.147177	1	native	4	sd
1	54586	0.902419	0.432384	2	native	4	sd
2	60109	0.952479	0.228479	3	native	4	sd
3	42936	1.009779	0.563210	4	native	4	sd
4	27497	1.079323	0.527191	5	native	4	sd
5	25522	1.120256	0.101704	6	native	4	sd
6	44751	1.140174	0.371456	7	native	4	sd,

feature_idx	entropy	activation_prob	rank	variant	layer	language
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0	11486	0.546145	0.147846	1	native	4	ur
1	66006	0.840774	0.747299	2	native	4	ur
2	126437	0.854812	0.159803	3	native	4	ur
3	63461	1.008931	0.343473	4	native	4	ur
4	73485	1.015106	0.251680	5	native	4	ur
5	21748	1.073274	0.178204	6	native	4	ur
6	42324	1.091412	0.401525	7	native	4	ur
7	71970	1.106871	0.107382	8	native	4	ur
8	92802	1.150200	0.273895	9	native	4	ur

9	76722	1.174336	0.103138	10	native	4	ur
10	57206	1.174989	0.600895	11	native	4	ur
11	120393	1.203785	0.436665	12	native	4	ur
12	130445	1.227692	0.100021	13	native	4	ur
13	41729	1.257743	0.393866	14	native	4	ur,
feature_idx entropy activation_prob rank variant layer language							
0	118853	0.340786	0.103403	1	native	4	bn
1	11770	0.454280	0.366985	2	native	4	bn
2	81393	0.464026	0.128307	3	native	4	bn
3	89150	0.477310	0.112091	4	native	4	bn
4	125956	0.497448	0.239193	5	native	4	bn
..	...	...	...	...	...	...	...
89	49092	1.280694	0.102955	90	native	4	bn
90	73131	1.291899	0.120512	91	native	4	bn
91	63519	1.292418	0.130575	92	native	4	bn
92	82905	1.306230	0.109015	93	native	4	bn
93	31219	1.311126	0.130783	94	native	4	bn

[94 rows x 7 columns],							
feature_idx entropy activation_prob rank variant layer language							
0	4032	0.217586	0.172918	1	native	5	hi
1	47139	0.253843	0.245701	2	native	5	hi
2	25324	0.315684	0.354477	3	native	5	hi
3	70326	0.371283	0.253641	4	native	5	hi
4	18441	0.426557	0.323414	5	native	5	hi
..	...	...	...	...	...	...	...
95	103069	0.896127	0.573345	96	native	5	hi
96	32317	0.898880	0.540304	97	native	5	hi
97	34391	0.903694	0.282614	98	native	5	hi
98	54215	0.903796	0.532001	99	native	5	hi
99	71842	0.915073	0.403891	100	native	5	hi

[100 rows x 7 columns],							
feature_idx entropy activation_prob rank variant layer language							
0	68328	0.678062	0.129011	1	native	5	sd
1	62076	0.685773	0.129606	2	native	5	sd
2	67347	0.785117	0.382380	3	native	5	sd
3	109519	0.916773	0.150454	4	native	5	sd
4	45976	0.933526	0.281490	5	native	5	sd
5	34486	0.976434	0.109770	6	native	5	sd
6	47326	0.980726	0.131237	7	native	5	sd
7	85640	1.002142	0.339033	8	native	5	sd
8	14949	1.010053	0.263659	9	native	5	sd
9	121803	1.027089	0.590580	10	native	5	sd
10	91310	1.050931	0.365904	11	native	5	sd
11	103502	1.094279	0.343611	12	native	5	sd
12	12253	1.128113	0.114584	13	native	5	sd
13	91477	1.142976	0.170237	14	native	5	sd
14	8725	1.225784	0.583017	15	native	5	sd,
feature_idx entropy activation_prob rank variant layer language							
0	22677	0.387299	0.187046	1	native	5	ur
1	47325	0.481401	0.126986	2	native	5	ur
2	126726	0.563512	0.177208	3	native	5	ur
3	113393	0.630242	0.107197	4	native	5	ur
4	41365	0.640313	0.494864	5	native	5	ur
5	49733	0.833318	0.348301	6	native	5	ur

6	112929	0.843538	0.189934	7	native	5	ur
7	11629	0.843618	0.151028	8	native	5	ur
8	58903	0.844520	0.165947	9	native	5	ur
9	58380	0.848580	0.385157	10	native	5	ur
10	48097	0.909324	0.288895	11	native	5	ur
11	78536	0.927214	0.138067	12	native	5	ur
12	63882	0.935247	0.292173	13	native	5	ur
13	72770	0.945868	0.196472	14	native	5	ur
14	38295	0.961393	0.375511	15	native	5	ur
15	121336	1.030108	0.401836	16	native	5	ur
16	98053	1.057003	0.404099	17	native	5	ur
17	8240	1.133220	0.140631	18	native	5	ur
18	27953	1.148708	0.439152	19	native	5	ur
19	72847	1.189703	0.401207	20	native	5	ur
20	9673	1.192837	0.332266	21	native	5	ur
21	112789	1.214396	0.537632	22	native	5	ur
22	1034	1.262368	0.279995	23	native	5	ur
23	83056	1.287630	0.434225	24	native	5	ur,
feature_idx entropy activation_prob rank variant layer language							
0	119850	0.138607	0.386526	1	native	5	bn
1	123915	0.339423	0.490011	2	native	5	bn
2	128075	0.365026	0.220501	3	native	5	bn
3	34876	0.366185	0.644064	4	native	5	bn
4	68687	0.393938	0.272562	5	native	5	bn
..	...	...	...	...	...	...	...
95	28288	1.156457	0.100174	96	native	5	bn
96	33625	1.161821	0.109459	97	native	5	bn
97	128889	1.173287	0.149502	98	native	5	bn
98	28012	1.176827	0.268415	99	native	5	bn
99	422	1.177024	0.529373	100	native	5	bn

[100 rows x 7 columns],							
feature_idx entropy activation_prob rank variant layer language							
0	17446	0.536685	0.341892	1	native	6	hi
1	121904	0.549588	0.144329	2	native	6	hi
2	42561	0.575698	0.193306	3	native	6	hi
3	85084	0.597224	0.321976	4	native	6	hi
4	88193	0.598651	0.194706	5	native	6	hi
..	...	...	...	...	...	...	...
95	115968	0.993513	0.560022	96	native	6	hi
96	54125	0.993948	0.398417	97	native	6	hi
97	125033	0.995222	0.740852	98	native	6	hi
98	96282	0.996239	0.324812	99	native	6	hi
99	47556	0.997496	0.391138	100	native	6	hi

[100 rows x 7 columns],							
feature_idx entropy activation_prob rank variant layer language							
0	102053	0.805024	0.191561	1	native	6	sd
1	76881	0.807733	0.373810	2	native	6	sd
2	72786	0.840375	0.294893	3	native	6	sd
3	6750	0.911501	0.474884	4	native	6	sd
4	87999	0.917313	0.537743	5	native	6	sd
5	110508	0.950261	0.341091	6	native	6	sd
6	71706	0.970090	0.610376	7	native	6	sd
7	85223	1.025431	0.268951	8	native	6	sd
8	51480	1.042448	0.246258	9	native	6	sd

9	81830	1.096233	0.213470	10	native	6	sd
10	50391	1.139283	0.109827	11	native	6	sd
11	62370	1.145042	0.544275	12	native	6	sd
12	17461	1.158640	0.292126	13	native	6	sd
13	13399	1.169497	0.278521	14	native	6	sd
14	121930	1.171793	0.360653	15	native	6	sd
15	118738	1.282131	0.655075	16	native	6	sd,

feature_idx	entropy	activation_prob	rank	variant	layer	language	
0	107504	0.450361	0.205923	1	native	6	ur
1	8310	0.695308	0.266270	2	native	6	ur
2	100886	0.747418	0.133747	3	native	6	ur
3	126999	0.761011	0.365063	4	native	6	ur
4	82694	0.798017	0.443033	5	native	6	ur
5	107794	0.806069	0.526176	6	native	6	ur
6	50392	0.833585	0.466904	7	native	6	ur
7	22260	0.833721	0.417309	8	native	6	ur
8	75969	0.872990	0.236614	9	native	6	ur
9	115356	0.883213	0.296482	10	native	6	ur
10	101476	0.892020	0.141315	11	native	6	ur
11	93855	0.927912	0.385569	12	native	6	ur
12	65821	0.929395	0.451830	13	native	6	ur
13	70319	0.975438	0.168770	14	native	6	ur
14	83212	0.983929	0.145056	15	native	6	ur
15	82826	0.987425	0.124527	16	native	6	ur
16	36496	0.993439	0.561665	17	native	6	ur
17	5368	0.994860	0.193346	18	native	6	ur
18	52963	0.996401	0.248339	19	native	6	ur
19	66172	1.009104	0.107182	20	native	6	ur
20	71418	1.023867	0.208353	21	native	6	ur
21	57227	1.043054	0.371759	22	native	6	ur
22	33007	1.057961	0.386602	23	native	6	ur
23	114935	1.064437	0.732767	24	native	6	ur
24	1776	1.079784	0.208975	25	native	6	ur
25	101160	1.080765	0.194160	26	native	6	ur
26	106212	1.085179	0.284607	27	native	6	ur
27	77972	1.090166	0.180641	28	native	6	ur
28	17862	1.130098	0.217358	29	native	6	ur
29	52619	1.153462	0.168579	30	native	6	ur
30	66013	1.184247	0.238405	31	native	6	ur
31	63633	1.214566	0.174318	32	native	6	ur
32	95192	1.241993	0.210615	33	native	6	ur
33	125866	1.306538	0.632353	34	native	6	ur,

feature_idx	entropy	activation_prob	rank	variant	layer	language	
0	60839	0.064536	0.173018	1	native	6	bn
1	100724	0.201433	0.420769	2	native	6	bn
2	18544	0.255511	0.366203	3	native	6	bn
3	118884	0.266163	0.301128	4	native	6	bn
4	14421	0.278066	0.534946	5	native	6	bn
..	...	...	...	...	...	...	...
95	66404	0.740246	0.734140	96	native	6	bn
96	111069	0.741122	0.193262	97	native	6	bn
97	126564	0.743509	0.202248	98	native	6	bn
98	34646	0.745615	0.459450	99	native	6	bn
99	86902	0.747310	0.231853	100	native	6	bn

[100 rows x 7 columns], feature_idx entropy activation_prob rank var

iant	layer	language					
0	127009	0.244622	0.250130	1	native	7	hi
1	71743	0.245703	0.150795	2	native	7	hi
2	108375	0.276613	0.308105	3	native	7	hi
3	6083	0.287308	0.108370	4	native	7	hi
4	114405	0.309442	0.351782	5	native	7	hi
..	...	...	...	...	...	...	...
95	104452	0.779739	0.162926	96	native	7	hi
96	80621	0.780829	0.422777	97	native	7	hi
97	90884	0.785244	0.198086	98	native	7	hi
98	26821	0.785833	0.695562	99	native	7	hi
99	35694	0.786900	0.273193	100	native	7	hi

[100 rows x 7 columns], feature_idx entropy activation_prob rank var

iant	layer	language					
0	37822	0.699503	0.204899	1	native	7	sd
1	14028	0.763669	0.144428	2	native	7	sd
2	106783	0.886693	0.330263	3	native	7	sd
3	3014	0.895475	0.167557	4	native	7	sd
4	90836	0.934217	0.139751	5	native	7	sd
5	11435	0.935874	0.368057	6	native	7	sd
6	24783	0.945791	0.219093	7	native	7	sd
7	77721	0.953008	0.324301	8	native	7	sd
8	33826	0.953010	0.272417	9	native	7	sd
9	85386	0.971163	0.220722	10	native	7	sd
10	118456	0.985836	0.158867	11	native	7	sd
11	4352	1.018175	0.735825	12	native	7	sd
12	26134	1.033083	0.729527	13	native	7	sd
13	79182	1.037084	0.300928	14	native	7	sd
14	39379	1.037328	0.406586	15	native	7	sd
15	27438	1.047428	0.622612	16	native	7	sd
16	33866	1.049815	0.225239	17	native	7	sd
17	78059	1.057164	0.210585	18	native	7	sd
18	117052	1.063967	0.184031	19	native	7	sd
19	110346	1.116714	0.542398	20	native	7	sd
20	60901	1.137241	0.192045	21	native	7	sd
21	75963	1.142375	0.437559	22	native	7	sd
22	112464	1.148696	0.129673	23	native	7	sd
23	5183	1.155447	0.514875	24	native	7	sd
24	36670	1.187155	0.602652	25	native	7	sd
25	52238	1.209732	0.251095	26	native	7	sd
26	23513	1.227116	0.357416	27	native	7	sd
27	3187	1.245585	0.110916	28	native	7	sd
28	108651	1.278269	0.309351	29	native	7	sd,

feature_idx entropy activation_prob rank variant layer language

0	16878	0.443546	0.107597	1	native	7	ur
1	39232	0.518938	0.410769	2	native	7	ur
2	126750	0.668657	0.310461	3	native	7	ur
3	123657	0.684976	0.191267	4	native	7	ur
4	123742	0.692506	0.207653	5	native	7	ur
5	60717	0.698941	0.346470	6	native	7	ur
6	46353	0.745804	0.285431	7	native	7	ur
7	26890	0.748761	0.287241	8	native	7	ur
8	68341	0.762169	0.451088	9	native	7	ur
9	120327	0.770643	0.204390	10	native	7	ur
10	5717	0.789753	0.267752	11	native	7	ur

11	3537	0.792443	0.127600	12	native	7	ur
12	114365	0.798538	0.236395	13	native	7	ur
13	59250	0.815817	0.419668	14	native	7	ur
14	66211	0.823209	0.376905	15	native	7	ur
15	70573	0.834369	0.233644	16	native	7	ur
16	74079	0.872732	0.481365	17	native	7	ur
17	17453	0.878551	0.371464	18	native	7	ur
18	113166	0.897240	0.256346	19	native	7	ur
19	3816	0.920202	0.100780	20	native	7	ur
20	36817	0.935304	0.349239	21	native	7	ur
21	72809	0.937575	0.530230	22	native	7	ur
22	15834	0.965368	0.154612	23	native	7	ur
23	44286	0.966974	0.157297	24	native	7	ur
24	120220	0.967280	0.155209	25	native	7	ur
25	100661	0.973997	0.396729	26	native	7	ur
26	95306	0.985352	0.261650	27	native	7	ur
27	72782	0.989851	0.438673	28	native	7	ur
28	63549	1.008201	0.482473	29	native	7	ur
29	88640	1.013553	0.494352	30	native	7	ur
30	9672	1.016655	0.220615	31	native	7	ur
31	26459	1.022770	0.416626	32	native	7	ur
32	102315	1.034605	0.250175	33	native	7	ur
33	130259	1.051662	0.472631	34	native	7	ur
34	58186	1.064400	0.315842	35	native	7	ur
35	78608	1.089559	0.427488	36	native	7	ur
36	110340	1.108333	0.346411	37	native	7	ur
37	88943	1.111096	0.208960	38	native	7	ur
38	7114	1.134008	0.257961	39	native	7	ur
39	21372	1.139600	0.153864	40	native	7	ur
40	100087	1.142952	0.224691	41	native	7	ur
41	125195	1.151898	0.170778	42	native	7	ur
42	29078	1.159902	0.326550	43	native	7	ur
43	107646	1.162760	0.461021	44	native	7	ur
44	96701	1.170419	0.268538	45	native	7	ur
45	114644	1.190818	0.614992	46	native	7	ur
46	77276	1.199320	0.106134	47	native	7	ur
47	102661	1.199696	0.225883	48	native	7	ur
48	57831	1.202903	0.716393	49	native	7	ur
49	84183	1.208757	0.668858	50	native	7	ur
50	94807	1.210254	0.625557	51	native	7	ur
51	130692	1.227327	0.106306	52	native	7	ur,

feature_idx	entropy	activation_prob	rank	variant	layer	language
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0	10445	0.115132	0.534714	1	native	7	bn
1	9559	0.172530	0.649775	2	native	7	bn
2	120313	0.178084	0.589148	3	native	7	bn
3	32350	0.180815	0.290201	4	native	7	bn
4	63385	0.225100	0.672256	5	native	7	bn
..	...	...	...	...	...	...	...
95	98198	0.793582	0.705303	96	native	7	bn
96	103402	0.794227	0.102604	97	native	7	bn
97	10949	0.795326	0.615532	98	native	7	bn
98	55974	0.801858	0.110098	99	native	7	bn
99	118046	0.807562	0.108353	100	native	7	bn

[100 rows x 7 columns], feature_idx entropy activation_prob rank variant layer language

0	95105	0.287682	0.117023	1	native	8	hi
1	70585	0.292696	0.106428	2	native	8	hi
2	94301	0.308818	0.253723	3	native	8	hi
3	81439	0.386021	0.474476	4	native	8	hi
4	42565	0.415311	0.167820	5	native	8	hi
..	...	...	...	...	...	...	...
95	29427	0.969757	0.431515	96	native	8	hi
96	55747	0.973929	0.395910	97	native	8	hi
97	127512	0.975547	0.394920	98	native	8	hi
98	107997	0.978886	0.791632	99	native	8	hi
99	37887	0.981602	0.655140	100	native	8	hi

[100 rows x 7 columns],	feature_idx	entropy	activation_prob	rank	variant	layer	language
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0	82124	0.645793	0.113132	1	native	8	sd
1	82686	0.660742	0.300192	2	native	8	sd
2	120590	0.730823	0.347992	3	native	8	sd
3	62940	0.765307	0.231215	4	native	8	sd
4	18513	0.783665	0.202015	5	native	8	sd
5	129042	0.934341	0.588556	6	native	8	sd
6	101791	1.023966	0.184696	7	native	8	sd
7	26044	1.087491	0.262802	8	native	8	sd
8	118249	1.097665	0.296272	9	native	8	sd
9	49322	1.128125	0.372732	10	native	8	sd
10	78538	1.167663	0.534863	11	native	8	sd
11	24441	1.208992	0.155222	12	native	8	sd
12	95639	1.260287	0.250790	13	native	8	sd,

feature_idx	entropy	activation_prob	rank	variant	layer	language
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0	103407	0.327939	0.249063	1	native	8	ur
1	57930	0.462347	0.141731	2	native	8	ur
2	18578	0.577233	0.369690	3	native	8	ur
3	93636	0.626865	0.115406	4	native	8	ur
4	13658	0.695858	0.263330	5	native	8	ur
5	12451	0.710570	0.637314	6	native	8	ur
6	2328	0.728716	0.191099	7	native	8	ur
7	40569	0.736577	0.190756	8	native	8	ur
8	10645	0.740267	0.114475	9	native	8	ur
9	46530	0.763785	0.280262	10	native	8	ur
10	64009	0.787410	0.270468	11	native	8	ur
11	78793	0.877249	0.115322	12	native	8	ur
12	62205	0.903774	0.134176	13	native	8	ur
13	100316	0.928964	0.188597	14	native	8	ur
14	59613	0.933209	0.634516	15	native	8	ur
15	26027	0.963963	0.603561	16	native	8	ur
16	12795	0.980659	0.432789	17	native	8	ur
17	8251	1.002723	0.178801	18	native	8	ur
18	93869	1.013022	0.409180	19	native	8	ur
19	97087	1.013526	0.176100	20	native	8	ur
20	6598	1.034477	0.678329	21	native	8	ur
21	91237	1.063398	0.596197	22	native	8	ur
22	104304	1.065149	0.262161	23	native	8	ur
23	8955	1.091457	0.263552	24	native	8	ur
24	25712	1.099951	0.152893	25	native	8	ur
25	90764	1.110582	0.104973	26	native	8	ur
26	5029	1.125925	0.121563	27	native	8	ur
27	67383	1.173284	0.146482	28	native	8	ur

28	93801	1.185195	0.268473	29	native	8	ur
29	70139	1.246124	0.225149	30	native	8	ur
30	78801	1.270269	0.142773	31	native	8	ur
31	65863	1.286036	0.111811	32	native	8	ur,
feature_idx entropy activation_prob rank variant layer language							
0	46515	0.307537	0.546928	1	native	8	bn
1	130928	0.365256	0.168512	2	native	8	bn
2	45221	0.436766	0.111017	3	native	8	bn
3	87763	0.447049	0.118967	4	native	8	bn
4	71216	0.460818	0.496729	5	native	8	bn
..	...	...	...	...	...	...	...
95	37346	1.046303	0.119459	96	native	8	bn
96	92332	1.056129	0.329933	97	native	8	bn
97	22106	1.059901	0.106926	98	native	8	bn
98	30085	1.060188	0.225294	99	native	8	bn
99	43269	1.062396	0.312050	100	native	8	bn

[100 rows x 7 columns],							
variant layer language							
feature_idx entropy activation_prob rank var							
0	61493	0.360224	0.223667	1	native	9	hi
1	63798	0.392165	0.388933	2	native	9	hi
2	62908	0.437122	0.152121	3	native	9	hi
3	54578	0.441578	0.328499	4	native	9	hi
4	101730	0.461004	0.266512	5	native	9	hi
..	...	...	...	...	...	...	...
95	66717	1.136756	0.354790	96	native	9	hi
96	111510	1.141590	0.442303	97	native	9	hi
97	97046	1.142946	0.448229	98	native	9	hi
98	74285	1.144998	0.107967	99	native	9	hi
99	93190	1.145647	0.536392	100	native	9	hi

[100 rows x 7 columns],							
variant layer language							
feature_idx entropy activation_prob rank var							
0	123229	0.608654	0.197311	1	native	9	sd
1	74376	0.724760	0.263445	2	native	9	sd
2	116401	0.829822	0.640194	3	native	9	sd
3	38337	0.833839	0.153450	4	native	9	sd
4	69469	0.841954	0.388235	5	native	9	sd
5	69627	0.858178	0.142345	6	native	9	sd
6	35730	0.878172	0.325960	7	native	9	sd
7	126879	0.909920	0.407785	8	native	9	sd
8	113354	0.916544	0.546217	9	native	9	sd
9	13438	0.926445	0.152954	10	native	9	sd
10	55420	0.930011	0.391895	11	native	9	sd
11	3979	0.941006	0.223541	12	native	9	sd
12	80357	0.950833	0.113050	13	native	9	sd
13	61723	0.956706	0.140579	14	native	9	sd
14	13895	0.970876	0.376617	15	native	9	sd
15	87486	0.971317	0.631464	16	native	9	sd
16	88553	0.979153	0.198399	17	native	9	sd
17	9339	0.984153	0.306522	18	native	9	sd
18	90646	1.001111	0.241493	19	native	9	sd
19	98126	1.069394	0.509636	20	native	9	sd
20	118392	1.079479	0.140247	21	native	9	sd
21	42677	1.104352	0.608580	22	native	9	sd
22	77043	1.109517	0.510811	23	native	9	sd

23	114855	1.120077	0.127882	24	native	9	sd
24	21423	1.138768	0.440466	25	native	9	sd
25	61656	1.174383	0.289865	26	native	9	sd
26	129111	1.247178	0.576155	27	native	9	sd
27	4430	1.247398	0.496910	28	native	9	sd
28	127174	1.264411	0.294245	29	native	9	sd
29	121854	1.270263	0.503426	30	native	9	sd
30	86237	1.319546	0.138138	31	native	9	sd,
feature_idx entropy activation_prob rank variant layer language							
0	11752	0.629756	0.284811	1	native	9	ur
1	21757	0.750072	0.150667	2	native	9	ur
2	41193	0.789260	0.328083	3	native	9	ur
3	51107	0.830868	0.415403	4	native	9	ur
4	110707	0.842732	0.368752	5	native	9	ur
5	43161	0.869220	0.424379	6	native	9	ur
6	100005	0.913949	0.470602	7	native	9	ur
7	83384	1.080458	0.635180	8	native	9	ur
8	52402	1.095746	0.464796	9	native	9	ur
9	21483	1.124329	0.490013	10	native	9	ur
10	49587	1.172938	0.192244	11	native	9	ur
11	87578	1.203773	0.103533	12	native	9	ur
12	128058	1.227383	0.214428	13	native	9	ur
13	47054	1.233137	0.170036	14	native	9	ur
14	82248	1.283735	0.101917	15	native	9	ur,
feature_idx entropy activation_prob rank variant layer language							
0	105205	0.130511	0.620911	1	native	9	bn
1	14634	0.367555	0.320236	2	native	9	bn
2	128315	0.383841	0.618185	3	native	9	bn
3	119625	0.560913	0.439518	4	native	9	bn
4	113107	0.608316	0.730923	5	native	9	bn
..	...	...	...	...	...	...	...
95	74967	1.304748	0.125490	96	native	9	bn
96	74610	1.305123	0.112791	97	native	9	bn
97	76577	1.305520	0.103935	98	native	9	bn
98	64863	1.308056	0.150412	99	native	9	bn
99	47113	1.311929	0.119259	100	native	9	bn

[100 rows x 7 columns],							
variant	layer	language	feature_idx	entropy	activation_prob	rank	var
0	88307	0.241521	0.152971	1	native	10	hi
1	126404	0.261781	0.160624	2	native	10	hi
2	104804	0.320593	0.149509	3	native	10	hi
3	60535	0.445323	0.115124	4	native	10	hi
4	113353	0.505494	0.238970	5	native	10	hi
..	...	...	...	...	...	...	...
95	94287	1.020058	0.123762	96	native	10	hi
96	5275	1.022594	0.376400	97	native	10	hi
97	26895	1.022986	0.677668	98	native	10	hi
98	88999	1.023033	0.301064	99	native	10	hi
99	58631	1.024551	0.288528	100	native	10	hi

[100 rows x 7 columns],							
variant	layer	language	feature_idx	entropy	activation_prob	rank	var
0	93613	0.751195	0.532440	1	native	10	sd
1	80967	0.792047	0.170916	2	native	10	sd
2	51243	0.813049	0.450006	3	native	10	sd

3	126164	0.819055	0.209738	4	native	10	sd
4	42928	0.917367	0.464216	5	native	10	sd
5	121556	0.936067	0.660900	6	native	10	sd
6	13860	0.948331	0.109175	7	native	10	sd
7	7152	0.975812	0.222767	8	native	10	sd
8	65988	1.007190	0.765160	9	native	10	sd
9	32704	1.013299	0.788975	10	native	10	sd
10	122120	1.037273	0.497618	11	native	10	sd
11	90344	1.070231	0.452251	12	native	10	sd
12	90804	1.072999	0.349045	13	native	10	sd
13	81022	1.114098	0.203123	14	native	10	sd
14	78614	1.145700	0.553215	15	native	10	sd
15	5710	1.157331	0.728434	16	native	10	sd
16	4542	1.212580	0.476231	17	native	10	sd
17	70316	1.228632	0.540779	18	native	10	sd
18	62661	1.231983	0.573658	19	native	10	sd
19	118550	1.286014	0.578400	20	native	10	sd
20	92920	1.308464	0.649054	21	native	10	sd,
feature_idx entropy activation_prob rank variant layer language							
0	79021	0.590797	0.390512	1	native	10	ur
1	51962	0.595602	0.386360	2	native	10	ur
2	101038	0.657552	0.191652	3	native	10	ur
3	3070	0.687200	0.144587	4	native	10	ur
4	109779	0.727153	0.105375	5	native	10	ur
5	121334	0.733472	0.281191	6	native	10	ur
6	124860	0.750823	0.650486	7	native	10	ur
7	128128	0.756113	0.302099	8	native	10	ur
8	52237	0.758740	0.659094	9	native	10	ur
9	24863	0.776504	0.396763	10	native	10	ur
10	44255	0.827293	0.385141	11	native	10	ur
11	116572	0.833284	0.376909	12	native	10	ur
12	57694	0.855701	0.524767	13	native	10	ur
13	42137	0.863132	0.254835	14	native	10	ur
14	34471	0.934498	0.287912	15	native	10	ur
15	33815	0.945242	0.320894	16	native	10	ur
16	58209	0.951857	0.803602	17	native	10	ur
17	58420	0.973509	0.126497	18	native	10	ur
18	39231	0.982903	0.315674	19	native	10	ur
19	8125	1.017000	0.159256	20	native	10	ur
20	57408	1.049035	0.151075	21	native	10	ur
21	101820	1.091968	0.150257	22	native	10	ur
22	97214	1.109541	0.205795	23	native	10	ur
23	61573	1.110243	0.159269	24	native	10	ur
24	93780	1.130967	0.166906	25	native	10	ur
25	15482	1.138028	0.516893	26	native	10	ur
26	29186	1.146943	0.545174	27	native	10	ur
27	4875	1.161267	0.109049	28	native	10	ur
28	7557	1.199413	0.130934	29	native	10	ur
29	81545	1.200909	0.104324	30	native	10	ur
30	28482	1.233426	0.151028	31	native	10	ur
31	45579	1.248367	0.171001	32	native	10	ur
32	108439	1.252497	0.142902	33	native	10	ur
33	82860	1.255207	0.165098	34	native	10	ur
34	98711	1.272817	0.391632	35	native	10	ur
35	54886	1.289495	0.130972	36	native	10	ur
36	80285	1.310730	0.508217	37	native	10	ur

37	68606	1.316700	0.550949	38	native	10	ur,
feature_idx	entropy	activation_prob	rank	variant	layer	language	
0	59389	0.161388	0.626919	1	native	10	bn
1	120013	0.177218	0.503796	2	native	10	bn
2	89912	0.250207	0.461046	3	native	10	bn
3	19487	0.271630	0.285940	4	native	10	bn
4	22877	0.277049	0.137423	5	native	10	bn
..	...	...	...	...	...	...	...
95	33826	1.017927	0.777136	96	native	10	bn
96	31540	1.022470	0.528461	97	native	10	bn
97	17702	1.025672	0.257299	98	native	10	bn
98	103300	1.027963	0.507847	99	native	10	bn
99	100987	1.037641	0.755014	100	native	10	bn

[100 rows x 7 columns],				feature_idx	entropy	activation_prob	rank	variant
layer language								
0	58797	0.176933	0.234520	1	native	11	hi	
1	94798	0.247413	0.233931	2	native	11	hi	
2	110916	0.267219	0.158310	3	native	11	hi	
3	106154	0.274694	0.245832	4	native	11	hi	
4	12505	0.300747	0.277353	5	native	11	hi	
..	...	...	...	...	...	...	...	...
95	79793	0.873495	0.485584	96	native	11	hi	
96	120323	0.884541	0.640795	97	native	11	hi	
97	121507	0.887995	0.139915	98	native	11	hi	
98	114908	0.888146	0.372021	99	native	11	hi	
99	123025	0.891947	0.244503	100	native	11	hi	

[100 rows x 7 columns],				feature_idx	entropy	activation_prob	rank	variant
layer language								
0	127934	0.560810	0.174768	1	native	11	sd	
1	29415	0.874438	0.259378	2	native	11	sd	
2	95067	0.944895	0.484579	3	native	11	sd	
3	28418	1.025589	0.426176	4	native	11	sd	
4	119400	1.046538	0.477745	5	native	11	sd	
5	120324	1.064310	0.310444	6	native	11	sd	
6	6730	1.068414	0.596447	7	native	11	sd	
7	89371	1.072724	0.338232	8	native	11	sd	
8	80173	1.145820	0.318262	9	native	11	sd	
9	122290	1.211630	0.522894	10	native	11	sd	
10	100825	1.227784	0.510553	11	native	11	sd	
11	26937	1.302615	0.241144	12	native	11	sd,	
feature_idx	entropy	activation_prob	rank	variant	layer	language		
0	92824	0.302455	0.103971	1	native	11	ur	
1	102107	0.413422	0.172382	2	native	11	ur	
2	45072	0.417631	0.140699	3	native	11	ur	
3	124259	0.512953	0.158079	4	native	11	ur	
4	45466	0.591103	0.399515	5	native	11	ur	
5	75289	0.609241	0.121649	6	native	11	ur	
6	12886	0.617169	0.273640	7	native	11	ur	
7	26027	0.670455	0.190340	8	native	11	ur	
8	40839	0.697628	0.125309	9	native	11	ur	
9	23310	0.700077	0.473194	10	native	11	ur	
10	13870	0.710040	0.648521	11	native	11	ur	
11	95537	0.746315	0.547953	12	native	11	ur	
12	130729	0.769142	0.250055	13	native	11	ur	

13	95156	0.805067	0.677588	14	native	11	ur
14	38033	0.808160	0.268848	15	native	11	ur
15	112047	0.808583	0.135542	16	native	11	ur
16	79319	0.822833	0.596939	17	native	11	ur
17	5921	0.831940	0.208103	18	native	11	ur
18	8994	0.854674	0.441303	19	native	11	ur
19	21953	0.860498	0.520353	20	native	11	ur
20	73023	0.879390	0.282356	21	native	11	ur
21	55903	0.882108	0.416693	22	native	11	ur
22	13676	0.896795	0.186161	23	native	11	ur
23	130701	0.911053	0.430494	24	native	11	ur
24	88067	0.943300	0.283436	25	native	11	ur
25	26575	0.944177	0.350763	26	native	11	ur
26	73906	0.977500	0.110394	27	native	11	ur
27	104675	0.981966	0.238575	28	native	11	ur
28	111958	0.985064	0.286615	29	native	11	ur
29	122599	0.985625	0.368288	30	native	11	ur
30	45086	0.995655	0.439981	31	native	11	ur
31	112854	1.002310	0.217714	32	native	11	ur
32	68152	1.013063	0.556828	33	native	11	ur
33	13520	1.024637	0.455898	34	native	11	ur
34	39721	1.067298	0.437941	35	native	11	ur
35	44126	1.089515	0.457150	36	native	11	ur
36	46101	1.090652	0.284073	37	native	11	ur
37	115853	1.120512	0.401556	38	native	11	ur
38	9968	1.125416	0.118309	39	native	11	ur
39	57327	1.126105	0.309982	40	native	11	ur
40	35373	1.133819	0.125114	41	native	11	ur
41	92192	1.149639	0.350147	42	native	11	ur
42	23192	1.165602	0.205347	43	native	11	ur
43	75755	1.168315	0.460615	44	native	11	ur
44	10051	1.168851	0.377584	45	native	11	ur
45	98913	1.185348	0.297043	46	native	11	ur
46	129973	1.194260	0.391951	47	native	11	ur
47	68526	1.203884	0.317053	48	native	11	ur
48	45574	1.212173	0.179064	49	native	11	ur
49	131052	1.238500	0.287956	50	native	11	ur
50	56272	1.269087	0.622627	51	native	11	ur,

feature_idx	entropy	activation_prob	rank	variant	layer	language	
0	67261	0.089749	0.164030	1	native	11	bn
1	25098	0.166364	0.289244	2	native	11	bn
2	28176	0.170144	0.542049	3	native	11	bn
3	23702	0.181190	0.292145	4	native	11	bn
4	69332	0.196614	0.237854	5	native	11	bn
..	...	...	...	...	...	...	...
95	73077	0.653172	0.545143	96	native	11	bn
96	16106	0.657904	0.613773	97	native	11	bn
97	54521	0.658093	0.299257	98	native	11	bn
98	11779	0.663042	0.268221	99	native	11	bn
99	126942	0.670753	0.300190	100	native	11	bn

[100 rows x 7 columns],				feature_idx	entropy	activation_prob	rank	variant	layer	language
0	81687	0.302307	0.156092	1	native	12	hi			
1	90837	0.308213	0.218586	2	native	12	hi			
2	9034	0.462300	0.190767	3	native	12	hi			

3	123118	0.636046	0.303753	4	native	12	hi
4	4781	0.636771	0.347107	5	native	12	hi
..	...	...	...	...	...	...	...
95	11740	1.134399	0.446503	96	native	12	hi
96	119571	1.134709	0.447510	97	native	12	hi
97	6186	1.137323	0.446884	98	native	12	hi
98	120555	1.138736	0.285486	99	native	12	hi
99	121811	1.138796	0.438118	100	native	12	hi

[100 rows x 7 columns], feature_idx entropy activation_prob rank variant layer language

0	47175	1.083773	0.321817	1	native	12	sd
1	106812	1.181344	0.414875	2	native	12	sd,

feature_idx entropy activation_prob rank variant layer language

0	34314	0.336195	0.280132	1	native	12	ur
1	121099	0.717466	0.608160	2	native	12	ur
2	11443	0.888802	0.424414	3	native	12	ur
3	35977	0.974853	0.649220	4	native	12	ur
4	64470	1.021368	0.343220	5	native	12	ur
5	58086	1.071070	0.578789	6	native	12	ur
6	46483	1.217020	0.787189	7	native	12	ur
7	61985	1.218456	0.208021	8	native	12	ur
8	52876	1.291152	0.218832	9	native	12	ur,

feature_idx entropy activation_prob rank variant layer language

0	67845	0.559479	0.341810	1	native	12	bn
1	110041	0.660086	0.259180	2	native	12	bn
2	89373	0.660421	0.143581	3	native	12	bn
3	130435	0.669846	0.419643	4	native	12	bn
4	17074	0.696286	0.161865	5	native	12	bn
5	2440	0.708544	0.403698	6	native	12	bn
6	77947	0.732435	0.239389	7	native	12	bn
7	108859	0.745084	0.175871	8	native	12	bn
8	69964	0.764918	0.126865	9	native	12	bn
9	93354	0.780275	0.185797	10	native	12	bn
10	99430	0.796610	0.442806	11	native	12	bn
11	125681	0.821785	0.356043	12	native	12	bn
12	18569	0.832211	0.157360	13	native	12	bn
13	126593	0.879281	0.167393	14	native	12	bn
14	109400	0.883799	0.163874	15	native	12	bn
15	96042	0.898639	0.370995	16	native	12	bn
16	116963	0.906056	0.242290	17	native	12	bn
17	30039	0.914862	0.381214	18	native	12	bn
18	62931	0.922124	0.263638	19	native	12	bn
19	44605	0.926049	0.733120	20	native	12	bn
20	51780	0.928087	0.172287	21	native	12	bn
21	99958	1.022109	0.430698	22	native	12	bn
22	87677	1.031885	0.571142	23	native	12	bn
23	37844	1.050444	0.122540	24	native	12	bn
24	53421	1.079025	0.548979	25	native	12	bn
25	51906	1.095971	0.546677	26	native	12	bn
26	124477	1.140967	0.136497	27	native	12	bn
27	2088	1.143389	0.150290	28	native	12	bn
28	2029	1.171626	0.121357	29	native	12	bn
29	107047	1.205135	0.143173	30	native	12	bn
30	50154	1.208624	0.261677	31	native	12	bn
31	29902	1.226398	0.342035	32	native	12	bn,

feature_idx	entropy	activation_prob	rank	variant	layer	language	
0	17535	0.478949	0.170988	1	native	13	hi
1	31632	0.528341	0.187696	2	native	13	hi
2	40379	0.534223	0.193819	3	native	13	hi
3	75200	0.542113	0.176958	4	native	13	hi
4	75374	0.581041	0.134613	5	native	13	hi
..	...	...	...	...	...	...	...
57	94204	1.226164	0.166821	58	native	13	hi
58	80269	1.252906	0.607582	59	native	13	hi
59	120581	1.258698	0.142864	60	native	13	hi
60	64805	1.266977	0.541483	61	native	13	hi
61	126233	1.287675	0.103708	62	native	13	hi

[62 rows x 7 columns], feature_idx entropy activation_prob rank varia
nt layer language

0	101492	0.910673	0.250774	1	native	13	sd
1	73338	1.252215	0.253017	2	native	13	sd,

feature_idx	entropy	activation_prob	rank	variant	layer	language	
0	82277	0.630004	0.363119	1	native	13	ur
1	11303	0.823517	0.261728	2	native	13	ur
2	36160	0.874833	0.475115	3	native	13	ur
3	25265	1.046269	0.251162	4	native	13	ur
4	115249	1.145033	0.121283	5	native	13	ur
5	117943	1.185891	0.507648	6	native	13	ur,

feature_idx	entropy	activation_prob	rank	variant	layer	language	
0	120470	0.100562	0.358427	1	native	13	bn
1	103214	0.110556	0.368982	2	native	13	bn
2	22650	0.159867	0.733921	3	native	13	bn
3	82433	0.176703	0.265450	4	native	13	bn
4	102791	0.176896	0.177736	5	native	13	bn
..	...	...	...	...	...	...	...
95	74091	0.634200	0.130299	96	native	13	bn
96	123681	0.634252	0.109539	97	native	13	bn
97	129089	0.635801	0.619146	98	native	13	bn
98	121153	0.637564	0.447378	99	native	13	bn
99	66262	0.639746	0.280380	100	native	13	bn

[100 rows x 7 columns], feature_idx entropy activation_prob rank var
iant layer language

0	93319	0.413439	0.152750	1	native	14	hi
1	31396	0.494020	0.148100	2	native	14	hi
2	122631	0.540788	0.106525	3	native	14	hi
3	56311	0.552064	0.334877	4	native	14	hi
4	87335	0.552554	0.744989	5	native	14	hi
..	...	...	...	...	...	...	...
95	106654	1.109387	0.154032	96	native	14	hi
96	112106	1.117272	0.145695	97	native	14	hi
97	33664	1.119976	0.195631	98	native	14	hi
98	18632	1.122731	0.362402	99	native	14	hi
99	46164	1.127246	0.435755	100	native	14	hi

[100 rows x 7 columns], feature_idx entropy activation_prob rank var
iant layer language

0	124424	0.397374	0.125715	1	native	14	sd
1	80083	0.831855	0.346664	2	native	14	sd
2	118728	0.849559	0.171066	3	native	14	sd

3	13227	0.914345	0.469572	4	native	14	sd
4	48363	0.951757	0.113306	5	native	14	sd
5	8727	0.956406	0.540096	6	native	14	sd
6	26876	0.989538	0.453106	7	native	14	sd
7	80247	0.995268	0.182878	8	native	14	sd
8	16949	1.004818	0.170422	9	native	14	sd
9	25983	1.019415	0.198206	10	native	14	sd
10	79885	1.032987	0.746437	11	native	14	sd
11	128411	1.036044	0.105640	12	native	14	sd
12	65255	1.125903	0.179789	13	native	14	sd
13	81822	1.167953	0.505720	14	native	14	sd
14	62381	1.169673	0.531193	15	native	14	sd
15	122829	1.246927	0.152515	16	native	14	sd
16	84925	1.247184	0.270369	17	native	14	sd
17	97356	1.266409	0.170616	18	native	14	sd
18	36466	1.266790	0.401346	19	native	14	sd
19	68004	1.308261	0.160551	20	native	14	sd,

	feature_idx	entropy	activation_prob	rank	variant	layer	language
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0	116413	0.535478	0.460991	1	native	14	ur
1	75696	0.545065	0.121029	2	native	14	ur
2	51359	0.654343	0.178436	3	native	14	ur
3	61288	0.659408	0.376837	4	native	14	ur
4	77841	0.692818	0.402769	5	native	14	ur
5	14117	0.693379	0.420645	6	native	14	ur
6	30492	0.739090	0.355240	7	native	14	ur
7	4529	0.764517	0.453878	8	native	14	ur
8	113755	0.828745	0.185530	9	native	14	ur
9	67068	0.900150	0.517435	10	native	14	ur
10	94623	0.980323	0.139938	11	native	14	ur
11	33772	1.006378	0.632183	12	native	14	ur
12	15168	1.024499	0.146078	13	native	14	ur
13	60868	1.060045	0.284019	14	native	14	ur
14	47326	1.061072	0.134678	15	native	14	ur
15	90888	1.063005	0.306210	16	native	14	ur
16	90680	1.073676	0.346128	17	native	14	ur
17	96415	1.085893	0.534210	18	native	14	ur
18	28462	1.100750	0.376255	19	native	14	ur
19	7053	1.101918	0.160551	20	native	14	ur
20	80321	1.121765	0.755236	21	native	14	ur
21	51963	1.122781	0.103617	22	native	14	ur
22	51302	1.130496	0.103952	23	native	14	ur
23	105805	1.159836	0.398380	24	native	14	ur
24	107701	1.175619	0.141903	25	native	14	ur
25	111244	1.178300	0.141777	26	native	14	ur
26	37731	1.186201	0.726297	27	native	14	ur
27	121787	1.209190	0.155188	28	native	14	ur
28	107288	1.210647	0.677263	29	native	14	ur
29	89576	1.228851	0.137039	30	native	14	ur
30	104913	1.268145	0.195864	31	native	14	ur
31	98851	1.286677	0.157030	32	native	14	ur,

	feature_idx	entropy	activation_prob	rank	variant	layer	language
--	-------------	---------	-----------------	------	---------	-------	----------

0	18521	0.114858	0.157995	1	native	14	bn
1	50516	0.201793	0.174479	2	native	14	bn
2	85094	0.202823	0.156498	3	native	14	bn
3	45439	0.215060	0.163494	4	native	14	bn
4	21121	0.247429	0.158327	5	native	14	bn

..	...	...	...	...	...	...	...
95	107068	0.747758	0.738440	96	native	14	bn
96	57529	0.751757	0.385181	97	native	14	bn
97	91611	0.755915	0.171673	98	native	14	bn
98	359	0.756392	0.678963	99	native	14	bn
99	115422	0.758749	0.684454	100	native	14	bn

[100 rows x 7 columns],	feature_idx	entropy	activation_prob	rank	var
variant layer language					

0	100144	0.740426	0.338015	1	native	15	hi
1	86589	0.791592	0.264290	2	native	15	hi
2	15283	0.820698	0.213249	3	native	15	hi
3	44430	0.857847	0.413305	4	native	15	hi
4	108004	0.919114	0.519995	5	native	15	hi
5	109837	0.941208	0.297796	6	native	15	hi
6	74317	0.974748	0.273811	7	native	15	hi
7	15481	1.020959	0.229620	8	native	15	hi
8	29267	1.029963	0.669712	9	native	15	hi
9	65259	1.048959	0.403778	10	native	15	hi
10	33088	1.066586	0.661402	11	native	15	hi
11	106162	1.081982	0.638037	12	native	15	hi
12	65773	1.127469	0.456490	13	native	15	hi
13	79934	1.130268	0.808167	14	native	15	hi
14	102240	1.154950	0.285875	15	native	15	hi
15	73467	1.157611	0.203030	16	native	15	hi
16	65689	1.187951	0.502981	17	native	15	hi
17	123416	1.194531	0.855051	18	native	15	hi
18	100311	1.204419	0.607479	19	native	15	hi
19	52254	1.204953	0.391048	20	native	15	hi
20	127161	1.206203	0.767872	21	native	15	hi
21	23892	1.218639	0.599661	22	native	15	hi
22	54469	1.239757	0.117714	23	native	15	hi
23	5577	1.264838	0.262871	24	native	15	hi
24	25714	1.282784	0.675343	25	native	15	hi,

feature_idx	entropy	activation_prob	rank	variant	layer	language
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0	94601	1.24385	0.685205	1	native	15	sd,	f
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feature_idx	entropy	activation_prob	rank	variant	layer	language
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0	125945	0.98112	0.245409	1	native	15	ur,
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feature_idx	entropy	activation_prob	rank	variant	layer	language
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0	110312	0.821208	0.186544	1	native	15	bn
1	72534	1.187816	0.101597	2	native	15	bn
2	93168	1.210266	0.325003	3	native	15	bn
3	75805	1.222122	0.116377	4	native	15	bn
4	95412	1.223444	0.102938	5	native	15	bn
5	23978	1.230963	0.188597	6	native	15	bn
6	11598	1.231534	0.113016	7	native	15	bn
7	95468	1.245982	0.113230	8	native	15	bn
8	92562	1.255555	0.105515	9	native	15	bn
9	120669	1.259255	0.134182	10	native	15	bn
10	51789	1.263575	0.175747	11	native	15	bn
11	11006	1.274105	0.117998	12	native	15	bn
12	10855	1.274552	0.115454	13	native	15	bn
13	50094	1.275728	0.159386	14	native	15	bn
14	108262	1.282728	0.103165	15	native	15	bn
15	1993	1.286120	0.126522	16	native	15	bn
16	81820	1.286746	0.116095	17	native	15	bn

17	69558	1.287942	0.143408	18	native	15	bn
18	36962	1.288720	0.111998	19	native	15	bn
19	36029	1.293137	0.130646	20	native	15	bn
20	77273	1.296449	0.153233	21	native	15	bn
21	57501	1.296784	0.124277	22	native	15	bn
22	73607	1.299276	0.113491	23	native	15	bn
23	59267	1.302495	0.176435	24	native	15	bn
24	126529	1.304790	0.187958	25	native	15	bn
25	45438	1.309692	0.115290	26	native	15	bn
26	13358	1.314310	0.174826	27	native	15	bn
27	130490	1.317912	0.179779	28	native	15	bn,
feature_idx entropy activation_prob rank				variant layer language			
0	130721	1.231083	0.109940	1	romanized	0	sd
1	35986	1.245613	0.353111	2	romanized	0	sd
2	126470	1.267289	0.113033	3	romanized	0	sd,
feature_idx entropy activation_prob rank				variant layer language			
0	124209	1.175587	0.120884	1	romanized	0	ur
1	73545	1.252441	0.163622	2	romanized	0	ur
2	98978	1.255928	0.158600	3	romanized	0	ur
3	57639	1.289057	0.115953	4	romanized	0	ur
4	66723	1.297242	0.118437	5	romanized	0	ur,
feature_idx entropy activation_prob rank				variant layer language			
0	76571	1.299738	0.220369	1	romanized	0	bn,
feature_idx entropy activation_prob rank				variant layer language			
0	105661	1.148494	0.132753	1	romanized	1	sd,
feature_idx entropy activation_prob rank				variant layer language			
0	2382	1.256221	0.299067	1	romanized	1	bn,
feature_idx entropy activation_prob rank				variant layer language			
0	105811	1.218724	0.141397	1	romanized	2	sd,
feature_idx entropy activation_prob rank				variant layer language			
0	1028	1.153829	0.115709	1	romanized	4	sd
1	30384	1.255921	0.195217	2	romanized	4	sd,
feature_idx entropy activation_prob rank				variant layer language			
0	110146	1.224434	0.107779	1	romanized	4	ur
1	56551	1.253462	0.111492	2	romanized	4	ur,
feature_idx entropy activation_prob rank				variant layer language			
0	45025	1.234088	0.101013	1	romanized	5	sd
1	30198	1.266937	0.269325	2	romanized	5	sd,
feature_idx entropy activation_prob rank				variant layer language			
0	43429	1.229118	0.104574	1	romanized	6	sd,
feature_idx entropy activation_prob rank				variant layer language			
0	130328	1.247702	0.347742	1	romanized	6	bn,
feature_idx entropy activation_prob rank				variant layer language			
0	119935	1.165310	0.124025	1	romanized	7	sd
1	54154	1.234290	0.164255	2	romanized	7	sd
2	21015	1.250724	0.118206	3	romanized	7	sd,
feature_idx entropy activation_prob rank				variant layer language			
0	827	1.285219	0.399790	1	romanized	7	bn
1	45382	1.295356	0.269762	2	romanized	7	bn
2	115854	1.310983	0.584940	3	romanized	7	bn,
feature_idx entropy activation_prob rank				variant layer language			
0	86682	1.146242	0.114645	1	romanized	8	sd
1	32443	1.255433	0.179714	2	romanized	8	sd
2	119100	1.292345	0.169138	3	romanized	8	sd,
feature_idx entropy activation_prob rank				variant layer language			
0	116401	1.282878	0.463554	1	romanized	9	bn

1	80474	1.315335	0.734299	2	romanized	9	bn,
feature_idx	entropy	activation_prob	rank	variant	layer	language	
0	74599	1.069244	0.187508	1	romanized	10	sd
1	37493	1.308289	0.231314	2	romanized	10	sd,
feature_idx	entropy	activation_prob	rank	variant	layer	language	
0	109779	1.134843	0.137804	1	romanized	10	ur,
feature_idx	entropy	activation_prob	rank	variant	layer	language	
0	13811	1.274303	0.224667	1	romanized	10	bn
1	47356	1.297246	0.240349	2	romanized	10	bn
2	110774	1.316236	0.208357	3	romanized	10	bn,
feature_idx	entropy	activation_prob	rank	variant	layer	language	
0	101304	1.272928	0.333155	1	romanized	11	bn
1	61455	1.295832	0.294790	2	romanized	11	bn,
feature_idx	entropy	activation_prob	rank	variant	layer	language	
0	55963	1.299686	0.257774	1	romanized	14	sd]

	feature_idx	entropy	activation_prob	rank	variant	layer	language
0	31608	0.189624	0.187590	1	native	0	hi
1	4073	0.194023	0.193167	2	native	0	hi
2	105283	0.290827	0.344704	3	native	0	hi
3	44340	0.363738	0.184050	4	native	0	hi
4	35032	0.372147	0.177662	5	native	0	hi

[INFO] Loaded 3806 rows across 16 layers.

```
In [8]: # --- SAE-LAPE Layerwise Trends: Entropy and Activation Probability ---

# Layerwise aggregation
df_layer_summary = (
    df_sae.groupby(["variant", "layer"])
    .agg({
        "entropy": ["mean", "std"],
        "activation_prob": ["mean", "std"]
    })
    .reset_index()
)
df_layer_summary.columns = ["variant", "layer", "entropy_mean", "entropy_std", "activation_mean", "activation_std"]
display(df_layer_summary.head())

# === Plot 1: Mean Entropy Across Layers ===
plt.figure(figsize=(10,6))
for variant, color in zip(["native", "romanized"], ["C0", "C1"]):
    subset = df_layer_summary[df_layer_summary.variant == variant]
    plt.plot(subset.layer, subset.entropy_mean, marker='o', label=f"{variant} Mean Entropy")
    plt.fill_between(
        subset.layer,
        subset.entropy_mean - subset.entropy_std,
        subset.entropy_mean + subset.entropy_std,
        alpha=0.2
    )
plt.title("Mean Entropy of SAE-LAPE Selected Neurons Across Layers")
plt.xlabel("Layer")
```

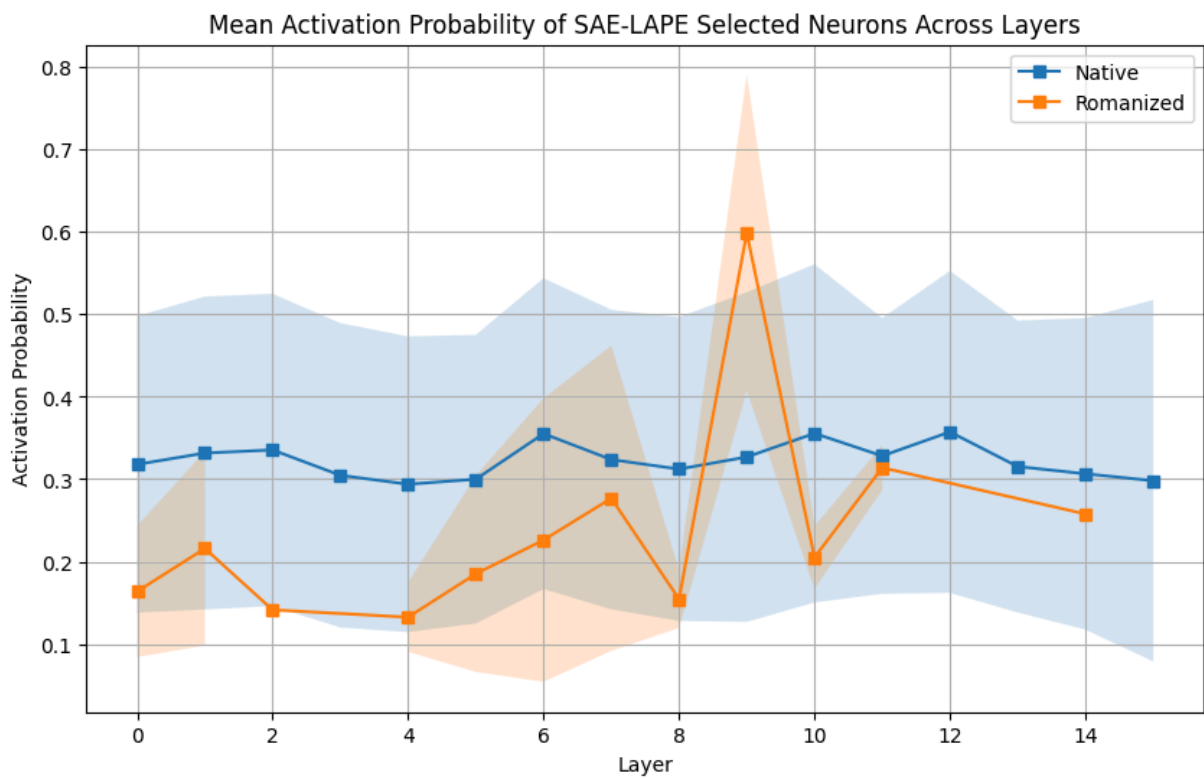
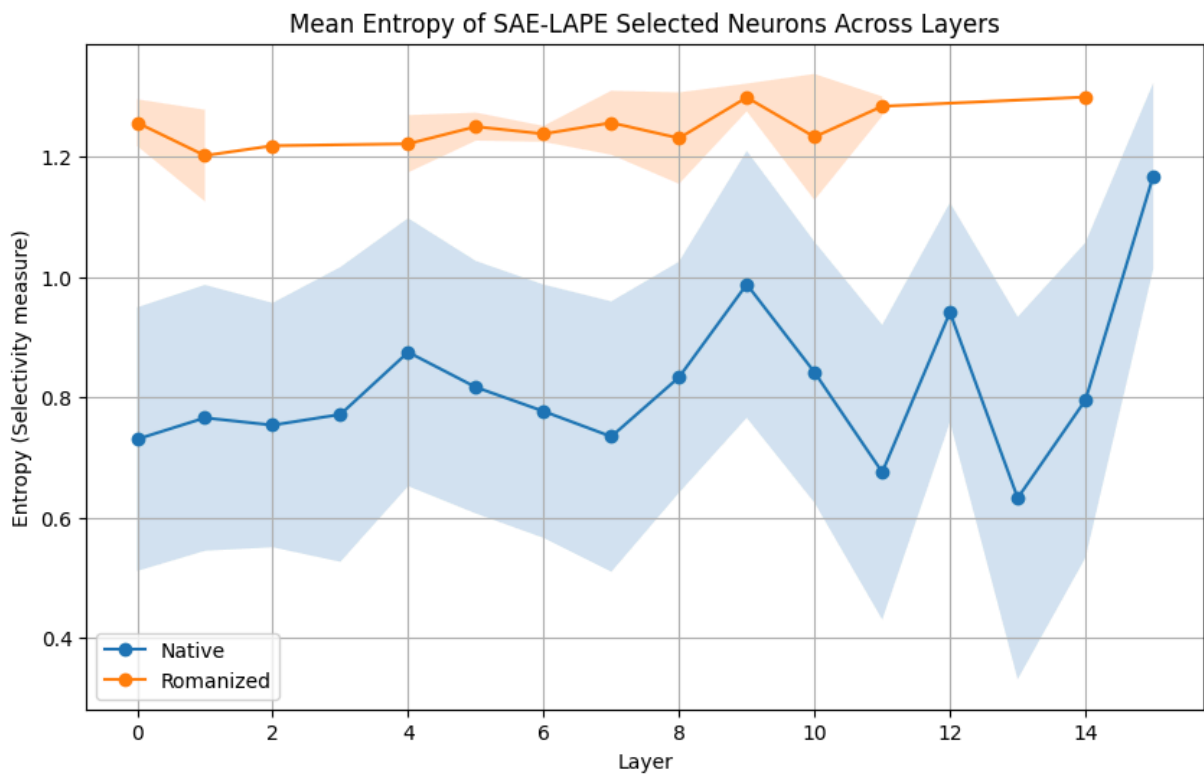
```

plt.ylabel("Entropy (Selectivity measure)")
plt.legend()
plt.grid(True)
plt.show()

# == Plot 2: Mean Activation Probability Across Layers ==
plt.figure(figsize=(10,6))
for variant, color in zip(["native", "romanized"], ["C2", "C3"]):
    subset = df_layer_summary[df_layer_summary.variant == variant]
    plt.plot(subset.layer, subset.act_mean, marker='s', label=f"{variant.capitalize()}")
    plt.fill_between(
        subset.layer,
        subset.act_mean - subset.act_std,
        subset.act_mean + subset.act_std,
        alpha=0.2
    )
plt.title("Mean Activation Probability of SAE-LAPE Selected Neurons Across Layers")
plt.xlabel("Layer")
plt.ylabel("Activation Probability")
plt.legend()
plt.grid(True)
plt.show()

```

	variant	layer	entropy_mean	entropy_std	act_mean	act_std
0	native	0	0.730459	0.219255	0.317965	0.179858
1	native	1	0.765976	0.221066	0.331730	0.189834
2	native	2	0.753860	0.203129	0.335498	0.189402
3	native	3	0.771597	0.244984	0.304696	0.184632
4	native	4	0.875237	0.222753	0.293887	0.179381



```
In [10]: # --- SAE-LAPE Per-Language Analysis ---
import seaborn as sns

df_lang_summary = (
    df_sae.groupby(["variant", "language"])
    .agg({
        "entropy": ["mean", "std"],
        "activation_prob": ["mean", "std"]
    })
)
```

```

    })
    .reset_index()
)
df_lang_summary.columns = ["variant", "language", "entropy_mean", "entropy_std"]

display(df_lang_summary)

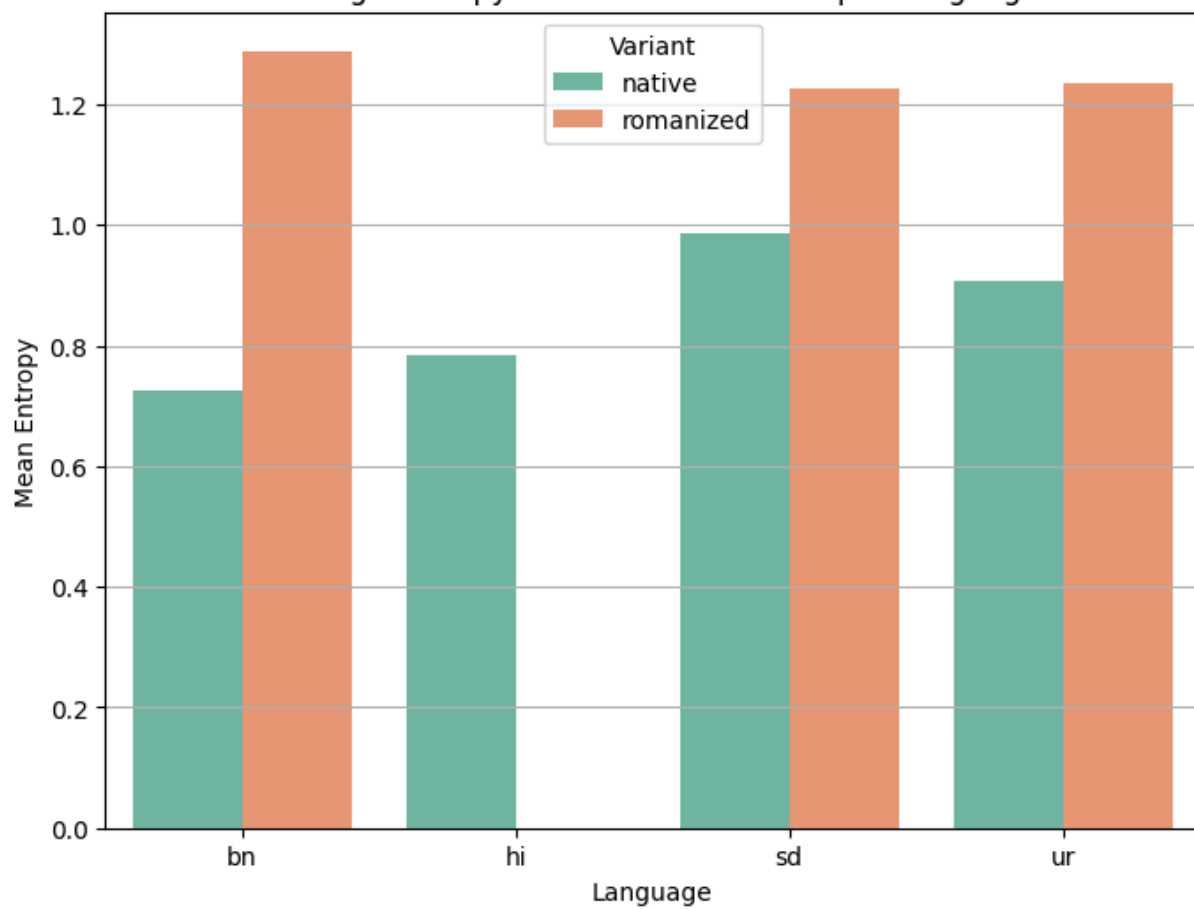
# === Plot: Entropy per Language ===
plt.figure(figsize=(8,6))
sns.barplot(
    x="language", y="entropy_mean", hue="variant",
    data=df_lang_summary, palette="Set2", capsize=0.2
)
plt.title("Average Entropy of SAE-LAPE Neurons per Language")
plt.ylabel("Mean Entropy")
plt.xlabel("Language")
plt.legend(title="Variant")
plt.grid(True, axis='y')
plt.show()

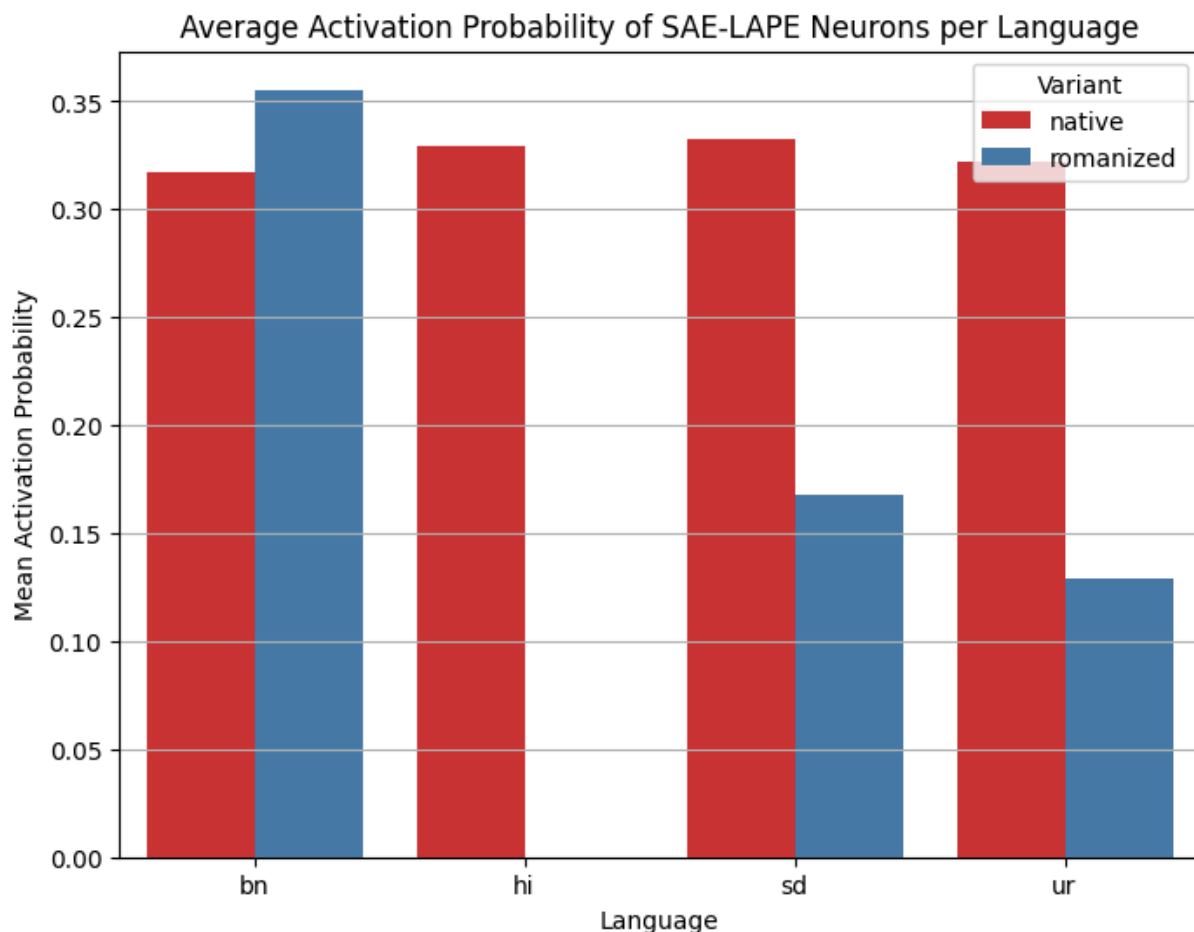
# === Plot: Activation Probability per Language ===
plt.figure(figsize=(8,6))
sns.barplot(
    x="language", y="act_mean", hue="variant",
    data=df_lang_summary, palette="Set1", capsize=0.2
)
plt.title("Average Activation Probability of SAE-LAPE Neurons per Language")
plt.ylabel("Mean Activation Probability")
plt.xlabel("Language")
plt.legend(title="Variant")
plt.grid(True, axis='y')
plt.show()

```

	variant	language	entropy_mean	entropy_std	act_mean	act_std
0	native	bn	0.726863	0.264286	0.316859	0.192722
1	native	hi	0.783336	0.195867	0.329575	0.186474
2	native	sd	0.986275	0.197499	0.332571	0.172368
3	native	ur	0.907105	0.236697	0.321890	0.180038
4	romanized	bn	1.288460	0.021392	0.355449	0.156748
5	romanized	sd	1.224877	0.061812	0.167508	0.068604
6	romanized	ur	1.235374	0.055486	0.129321	0.021567

Average Entropy of SAE-LAPE Neurons per Language





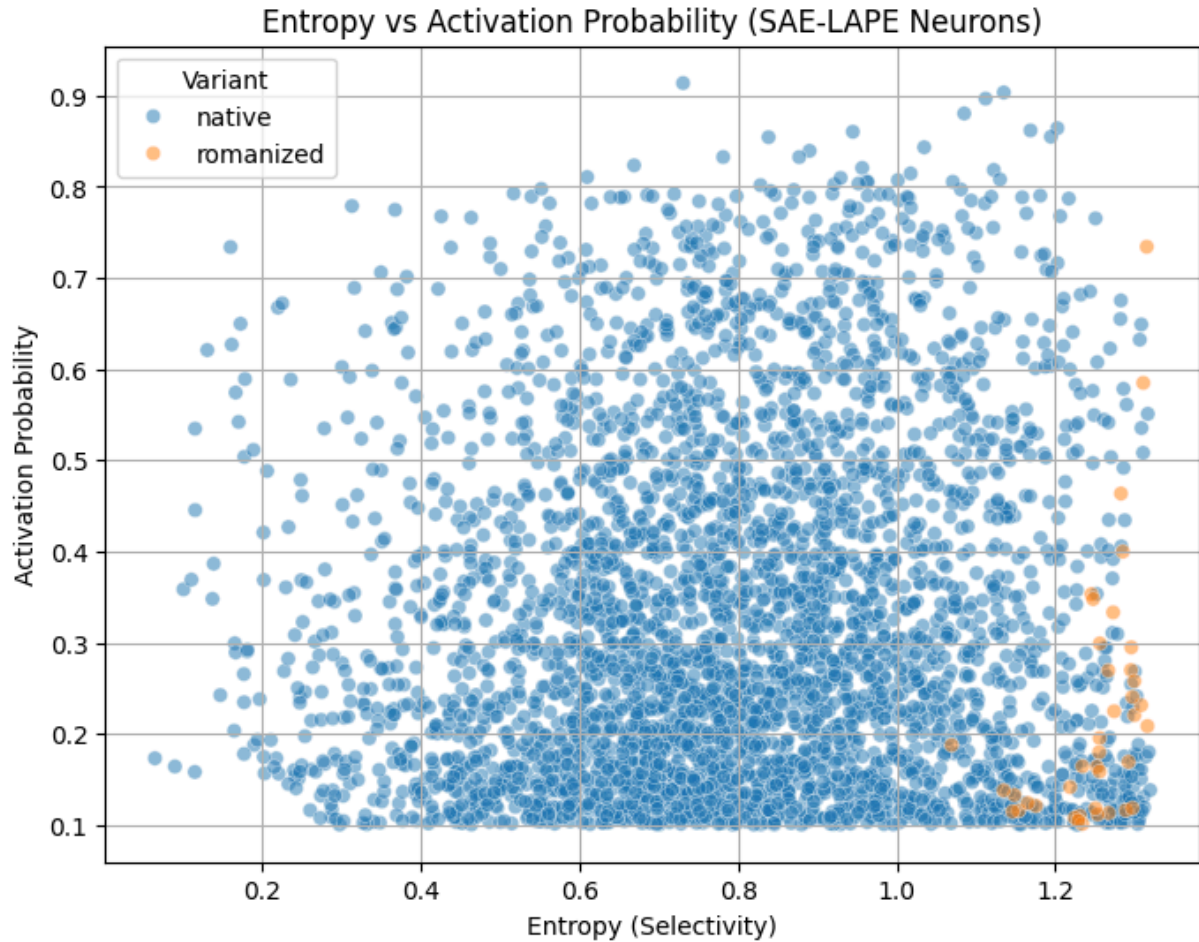
```
In [11]: # --- Entropy vs Activation Relationship ---

import seaborn as sns

plt.figure(figsize=(8,6))
sns.scatterplot(
    data=df_sae,
    x="entropy",
    y="activation_prob",
    hue="variant",
    alpha=0.5,
)
plt.title("Entropy vs Activation Probability (SAE-LAPE Neurons)")
plt.xlabel("Entropy (Selectivity)")
plt.ylabel("Activation Probability")
plt.legend(title="Variant")
plt.grid(True)
plt.show()

# Optionally, compute correlations per variant
corrs = (
    df_sae.groupby("variant")[["entropy", "activation_prob"]]
    .corr()
    .iloc[0::2,-1]
    .reset_index()
)
```

```
corrs.columns = ["variant", "metric", "correlation"]
display(corrs)
```



	variant	metric	correlation
0	native	entropy	0.059602
1	romanized	entropy	0.457192

```
In [ ]: # --- SAE-LAPE Selectivity Depth Profile (Entropy Heatmap) ---

import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np

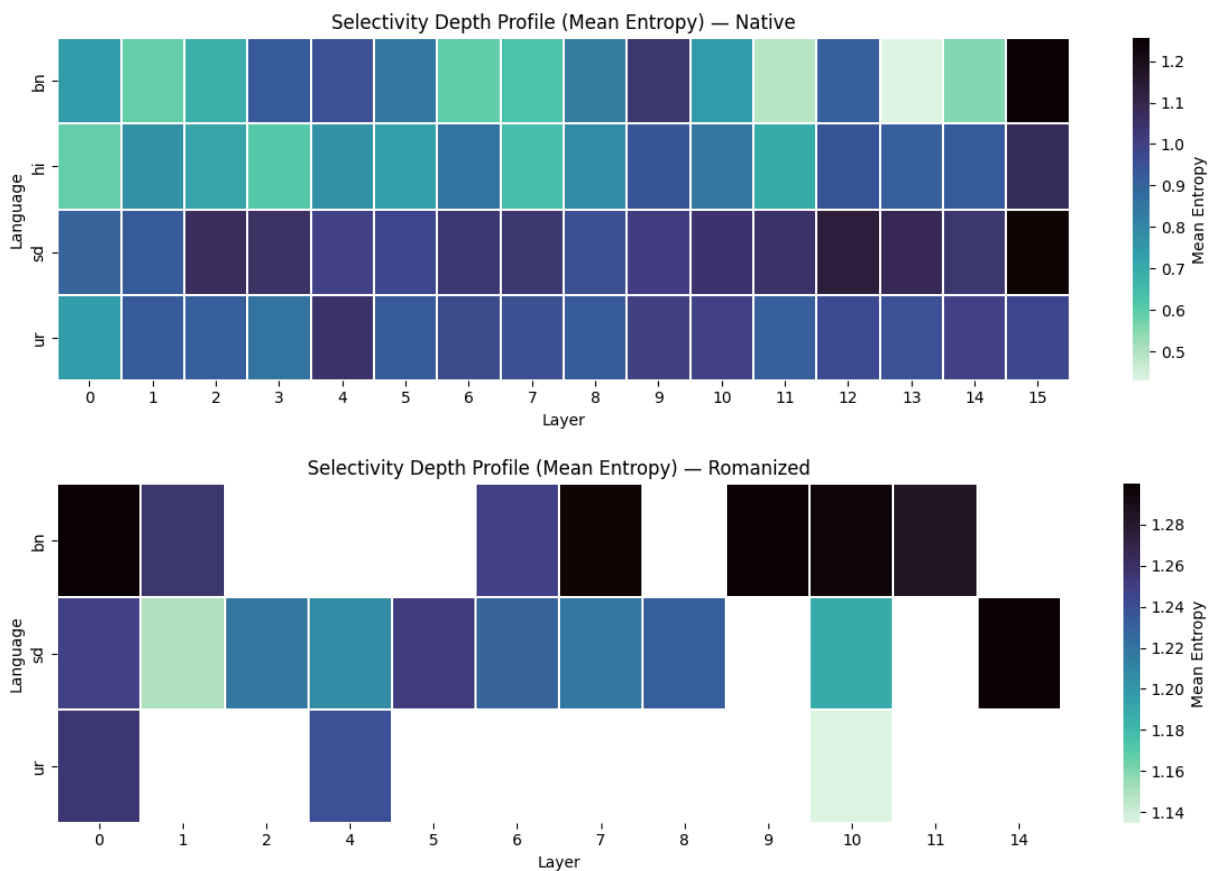
# Aggregate mean entropy per layer x language x variant
df_entropy = (
    df_sae.groupby(["variant", "language", "layer"])
    .agg(entropy_mean=("entropy", "mean"))
    .reset_index()
)

# Separate heatmaps for native and romanized
for variant in ["native", "romanized"]:
    subset = df_entropy[df_entropy.variant == variant]
    pivot = subset.pivot(index="language", columns="layer", values="entropy_
```

```

plt.figure(figsize=(12, 4))
sns.heatmap(
    pivot,
    cmap="mako_r",
    cbar_kws={'label': 'Mean Entropy'},
    vmin=pivot.min().min(),
    vmax=pivot.max().max(),
    linewidths=0.3,
    annot=False,
)
plt.title(f"Selectivity Depth Profile (Opp of Mean Entropy) – {variant.c}")
plt.xlabel("Layer")
plt.ylabel("Language")
plt.tight_layout()
plt.show()

```



In []: # --- SAE-LAPE Neuron Counts Across Layers and Languages ---

```

# Overall counts per variant × layer
df_counts_layer = (
    df_sae.groupby(["variant", "layer"])
    .size()
    .reset_index(name="num_neurons")
)

# Per-language counts per layer × variant
df_counts_lang = (
    df_sae.groupby(["variant", "layer", "language"])

```

```

        .size()
        .reset_index(name="num_neurons")
    )

display(df_counts_layer.head())
display(df_counts_lang.head())

# === Total neurons per layer ===
plt.figure(figsize=(10,6))
for variant, color in zip(["native", "romanized"], ["C0", "C1"]):
    subset = df_counts_layer[df_counts_layer.variant == variant]
    plt.plot(subset.layer, subset.num_neurons, marker='o', label=f"{variant}.")
plt.title("Total SAE-LAPE Neurons Identified per Layer")
plt.xlabel("Layer")
plt.ylabel("Neuron Count")
plt.legend()
plt.grid(True)
plt.show()

# === Per-language neuron counts per layer ===
g = sns.FacetGrid(df_counts_lang, col="variant", height=4, aspect=1.4, sharex=True)
g.map_dataframe(sns.lineplot, x="layer", y="num_neurons", hue="language", markers=True)
g.add_legend()
g.set_titles("Variant: {col_name}")
g.set_axis_labels("Layer", "Neuron Count")
g.fig.suptitle("SAE-LAPE Neuron Counts per Layer × Language", fontsize=15, y=1.05)
plt.show()

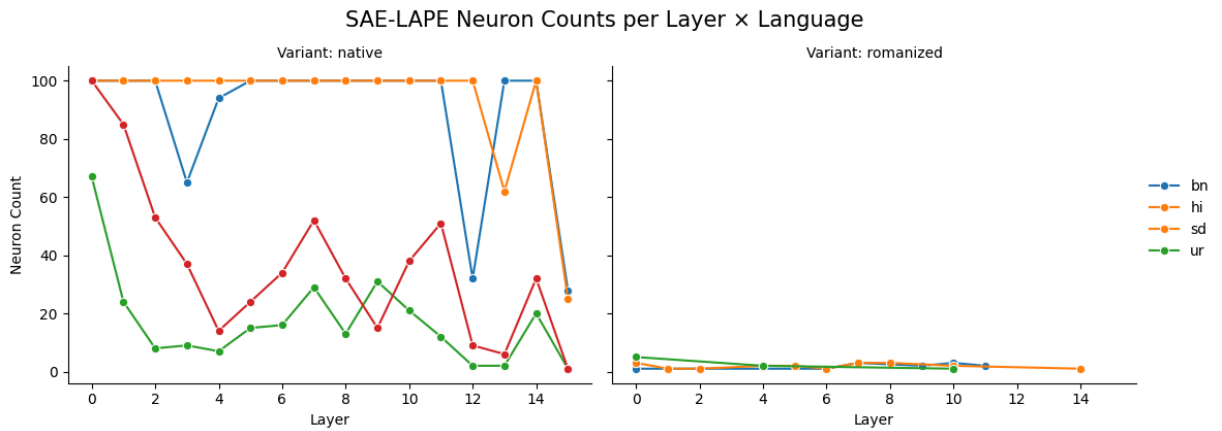
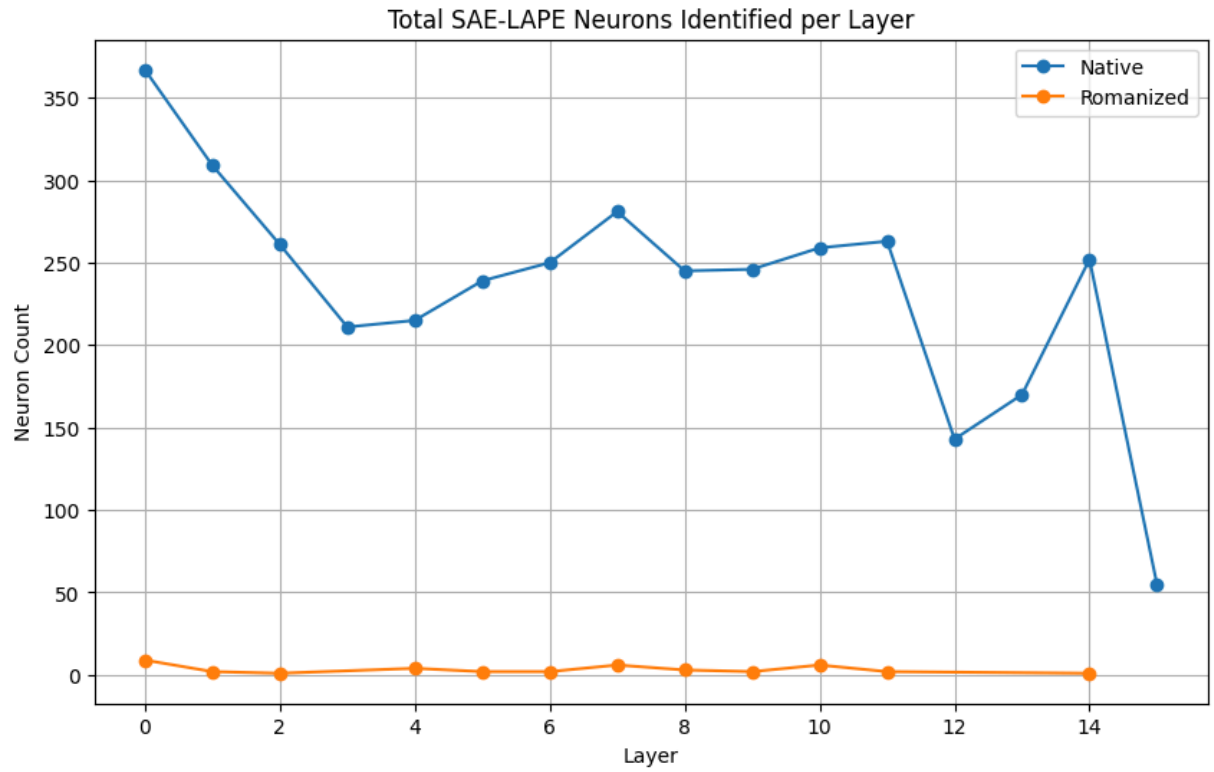
# === Average neurons per language across layers ===
avg_counts = (
    df_counts_lang.groupby(["variant", "language"])["num_neurons"]
    .mean()
    .reset_index()
)

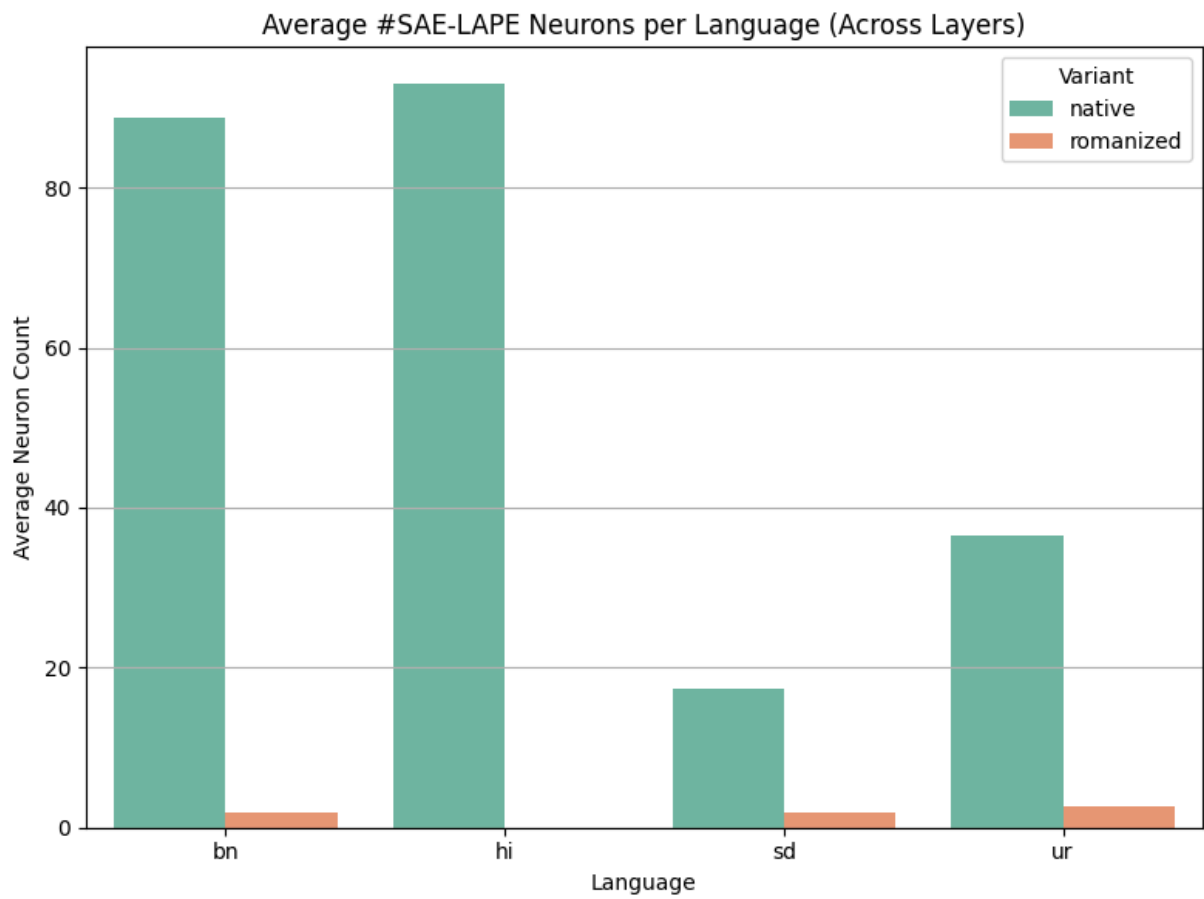
plt.figure(figsize=(8,6))
sns.barplot(x="language", y="num_neurons", hue="variant", data=avg_counts, palette="magma")
plt.title("Average #SAE-LAPE Neurons per Language (Across Layers)")
plt.ylabel("Average Neuron Count")
plt.xlabel("Language")
plt.legend(title="Variant")
plt.grid(True, axis="y")
plt.tight_layout()
plt.show()

```

	variant	layer	num_neurons
0	native	0	367
1	native	1	309
2	native	2	261
3	native	3	211
4	native	4	215

	variant	layer	language	num_neurons
0	native	0	bn	100
1	native	0	hi	100
2	native	0	sd	67
3	native	0	ur	100
4	native	1	bn	100





In []: